Public Transport (2026) 18:483–523
https://doi.org/10.1007/s12469-025-00404-y

ORIGINAL RESEARCH

Service design of shared first- and last-mile transit systems

Oksana Sabinik Pecherski[1] · **Tal Raviv**[2] · **Hillel Bar-Gera**[3]

Accepted: 23 July 2025 / Published online: 8 October 2025

Abstract
This paper explores the design and optimization of shared first- and last-mile transit systems (SFLMTS) as an essential component of public transport systems. It addresses the gap between the demand for efficient connectivity from suburban and rural areas to main transit hubs and the current underperformance of traditional public transportation systems in providing comprehensive last-mile solutions in sparsely populated areas. By focusing on demand-responsive transit (DRT) services, this study presents a novel framework for designing SFLMTS that leverages mobility-on-demand (MoD) principles to enhance the accessibility and efficiency of regional and national rail networks. The paper introduces a comprehensive framework that encompasses design and operational decisions, including passenger commitment, trip consolidation, and service-level considerations, which are crucial for optimizing the performance and economic viability of DRT services. Through a detailed case study, the paper illustrates the application of this framework using realistic data and simplified operational models (SOMs), demonstrating the potential of DRT to significantly reduce the number of required vehicles, decrease operational costs, and improve service levels compared to traditional fixed-route services. The findings highlight the importance of strategic design decisions in maximizing the efficiency and sustainability of first- and last-mile transit solutions, offering valuable insights for transportation planners, policymakers, and researchers.

Keywords Last-mile connectivity · Feeder transport service · Demand-responsive transit (DRT) · Mobility-on-demand (MoD)

✉ Oksana Sabinik Pecherski
pecho@sce.ac.il

1 Department of Industrial Engineering and Management, Sami Shamoon College of Engineering, Ashdod, Israel

2 School of Industrial & Intelligent Systems Engineering, Tel Aviv University, Tel-Aviv, Israel

3 Department of Industrial Engineering and Management, Ben-Gurion University of the Negev, Beer-Sheva, Israel

1 Introduction

Improving the viability of public transportation is vital for developing transportation systems. Effective public transport requires adequate connectivity between transit stations and trip origins/destinations (Lu et al. 2024). In particular, demand-responsive transit (DRT), also known as mobility-on-demand (MoD), can contribute to the sustainability of transportation systems (e.g., Yang and Lai 2020). In this paper, we study the design of train-feeder services by shared first- and last-mile transit systems (SFLMTS) utilizing DRT. A key challenge in service (or product) design is to find the right balance between economic viability and the market's requirements and expectations. Our primary motivation is the potential fit between DRT possibilities and SFLMTS needs, especially in the context of a feeder service for suburban/rural train stations.

Regional and national rail networks provide fast transportation that covers long distances efficiently, bringing many passengers from one point to another. Yet, these networks cannot offer a complete solution that includes first- and last-mile connectivity (Shaheen and Chan 2016). As a result, travelers may avoid the train service or travel to the train station by car, consuming fuel, creating pollution and traffic congestion, incurring parking costs, and overcrowding the parking lots of the train stations. Train stations in dense urban areas can be served effectively by local busses and other shared modes, but these solutions are less effective in suburban and rural areas. These observations suggest that modern technology-enabled DRT may fill the market gap of SFLMTS, thus justifying the development of a theoretical framework for the design and operation of such services.

DRT/MoD solutions have been studied outside the contexts of suburban/rural SFLMTS. In dense urban areas, the conventional public transport (PT) may often be more effective than DRT, whether in SFLMTS or as a general point-to-point service. On the other hand, in suburban areas with low demand, conventional PT is less effective. For example, Mehran et al. (2020) and Mishra and Mehran (2023) examined semi-flexible transit solutions. Navidi et al. (2018) examined the potential of point-to-point DRT for low-demand areas and showed that in certain cases, the level of service (LOS) can be improved by 200% at the same cost, but in other cases, waiting time may increase considerably. Indeed, in low-density suburban/rural areas, a point-to-point service by DRT may also be challenging due to the issues of trip consolidation. We believe that the niche of rural/suburban SFLMTS is a particularly viable application of DRT because of the inherent advantages of temporal and spatial trip consolidation, as many passengers disembark (or embark) a train at the same time and in the same place (see the detailed discussion in Sect. 3.2). The bi-directional match between need and solution is the basis for our focus on this specific context.

Previous studies of DRT for SFLMTS examined alternative optimization models, mainly from an operational perspective (see the details in Sect. 2). Each of these studies assumed a different service design, usually with limited discussion of design choices. In contrast, the literature on PT devotes substantial attention to system design, in addition to studies that focus on operations. For example, Corvello et al. (2023) examine the problem of designing optimal parking lots for public transport organizations, focusing on a real-world application in Southern Italy. Their primary objective is to balance cost efficiency and service levels in the public transport system by optimizing the number, size, and layout of these lots. The study emphasizes the importance of decision-support systems for public transport organizations at the strategic planning stage, by investigating aspects of customer satisfaction.

Ceder (2015, p. 5) divides PT planning decisions into four main categories: network route design, timetable development, vehicle scheduling, and crew scheduling. The first two categories are typically considered part of the service design, while the last two are part of the service operation. In some regions, PT service design and operation are divided institutionally between separate entities. In Israel, for example, PT design is done by the Israeli public transport authority (a national governmental agency), while operational decisions are made by various private transport companies, selected once or twice a decade in a tender process to operate clusters of bus routes.

In this study, we examine the service-design perspective of DRT. We consider the design problem of a regulator that contemplates whether to support (or allow) SFLMTS at various train stations and under which conditions. In practice, a coalition of stakeholders may be involved in the service-design process, but to simplify the discussion, we will assume that a single "regulator" represents all stakeholders. We present a design framework consisting of high-level and detailed choices (see Sect. 3). We also explore the role of qualitative arguments and quantitative evaluation for the various design choices.

Existing literature on DRT and SFLMTS offers a good starting point, especially in identifying part of the design attributes. For example, Markov et al. (2021) classify transportation services according to seven dimensions (see Sect. 2). Following their terminology, part of our study could be described as comparing two options of station-based time-based services with professional ridesharing: a) fixed lines without reservations (but possibly with registration); and b) demand-responsive routes with prebooked reservations. This description misses three detailed design choices. The most important one is station choice within station-based services. Station choice is a well-known issue within PT design but is not as thoroughly discussed in DRT and SFLMTS studies. We present several two-stage approaches for station choice (see Sect. 3.5). The second issue is market structure, and especially the impact of the number of operators on system performance. A third minor deviation from the classification of Markov et al. (2021) regards the conditions that enable a service to be time-based, or semi-time-based, while routes are demand-responsive (see Sect. 3.4).

While detailed design attributes are important, we argue that in the context of DRT, particularly for SFLMTS, the design process should begin with three high-level choices: commitment, trip consolidation, and service level. None of these are included in the classification of Markov et al. (2021). The latter two have been discussed to some extent in various DRT/SFLMTS studies (see Sect. 2). The issue of operator commitment is related to the issue of reliability, which has been considered extensively in the literature on PT (e.g., Godachevich and Tirachini 2021), as well as in some DRT studies (e.g., Bansal et al. 2019); however, especially in the context of DRT, there are important commitment aspects, both from the operator side as well as from the passenger side, that have not been - to the best of our knowledge - fully discussed. The design framework we propose is centered around these three high-level choices, their interplay, as well as their implications on various detailed design choices (see Sect. 3).

Based on a qualitative analysis of the high-level choices, we suggest a specific SFLMTS design with a shuttle service schedule synchronized with the train schedule. Shortly after a train arrives at a station, all interested disembarking passengers, considered as one batch, are served by shuttles traveling from the train station to bus stops near the passengers' destination. In the opposite direction, shuttles pick up passengers from bus stops near their origins and bring them to the train station shortly before the batch boards an outgoing train. The operator guarantees service to all passenger trip requests while respecting a predetermined service level in terms of maximum walking time to and from the pickup location and *maximum excess ride time*, which is the difference between the total ride time in the shared vehicle and the direct travel time between the passenger pickup and drop-off locations. In addition, the maximum number of stops along each vehicle's route is limited.

A key decision for the regulator is whether vehicle routes should be fixed or demand-responsive. With fixed lines, when the capacity of the vehicles is not binding, reservations are not needed, and all registered passenger addresses (or all potential addresses within a defined region) should be covered at each departure time. If routes are demand-responsive, each passenger needs to submit a service request in advance through a prebooking reservation system. The request specifies the passenger's desired destination (resp., origin). Once all requests in a batch have been submitted, the operator chooses vehicle routes and assigns passengers to them. The passengers are informed about the specific vehicle they should board and the bus stop where they will be dropped off (resp., picked up). Note that several passengers may be served at the same bus stop, if their final destinations (resp., origins) are within a predefined walking distance from the stop.

In addition to the choice between fixed lines and demand-responsive vehicle routes, the regulator should also determine whether to limit the number of operators in a single train station through a tender process. In such a tender, the regulator should also choose specific values for the LOS parameters. This part of the design process requires a quantitative assessment of the impacts of the various choices on costs and, thus, on service price.

Cost assessment at the design stage is challenging due to market price and demand uncertainties. The cost is also sensitive to operation implementation details, particularly fleet acquisition decisions. To reduce the sensitivity of the analysis to specific market prices, we focus on estimating three major cost components: fleet size (number of vehicles), vehicle capacity, and mean total vehicle travel times. Note that our analysis assumes that service is provided to all passenger requests (without rejections); therefore, the revenue is constant, and thus, minimizing cost is equivalent to maximizing profit.

To handle demand uncertainty, the designer will need to collect demand data and predict/estimate the batch sizes and their spatial and temporal distribution.

For illustrative purposes, our numerical results consider four hypothetical demand levels (20, 30, 40, or 50 passengers per batch). For each possible batch size, we consider a set of scenarios based on actual destinations collected in a passenger survey.

To deal with the uncertainty regarding detailed design choices possibly made by the operator, policymakers may prefer to rely on simplified operational models (SOMs) to estimate the operator's costs in different scenarios. In this study, we focus on analyses of three SOMs, as described in Table 1. While these SOMs focus on the operator's routing decisions, we will also discuss how to use their results to analyze other choices (mainly fleet acquisition), as well as the impact of the limitations of these SOMs.

The three SOMs we consider minimize total vehicle travel time while ensuring a specified LOS for all passengers. The SOMs differ in two aspects: whether the fleet size has a known limit (SOM2) or not (SOM1 and SOM3) and whether the operator offers each passenger several vehicle options with a stop within a valid walking distance from his destination (SOM1 and SOM2) or only one valid option (SOM3). The latter (SOM3) represents an operator that deals with capacity limitations by combining several strategies, including minimizing the maximal occupancy as a secondary objective. Indeed, none of the proposed SOMs include explicit constraints on vehicle capacity, as is typical in operational models. Yet, we will demonstrate the possibilities of using the results when evaluating vehicle size choices and determining whether capacity-constrained SOMs are needed as part of the design process. All three models assume that train frequency is low, and therefore, the service design can be performed for each train separately. Furthermore, we assume that in suburban areas, some trains are characterized by inbound passengers only (probably during the morning), while other trains are characterized by outbound passengers only (probably during the afternoon and evening).

We present a mathematical formulation for each of the three SOMs and discuss solution methods that combine complete route enumeration, set covering, and matching in a bi-partite graph (see Sect. 4). In these mathematical models, the representation of inbound and outbound passengers is equivalent, and, therefore, the exposition is restricted to the outbound case. There are practical differences between inbound and outbound passengers, such as the required booking lead time, but these are not considered in our simplified models.

We explore these models numerically using realistic data from a specific case study, which includes a specific, realistic network (a set of bus stops and travel

Table 1 Proposed simplified operational models for cost estimation in SFLMTS service design

Simplified operational model	Fleet size	Bus stop choice
SOM1: DRT 1-M	Unlimited	Passenger preference
SOM2: DRT 1-M LFS[1]	Limited	Passenger preference
SOM3: DRT 1-M MMO[2]	Unlimited	Min–max occupancy

[1]Limited fleet size; [2]min–max occupancy

times between them) and the geographic distribution of demand (based on a survey of train passengers). Our numerical results show that all three SOMs can be solved optimally within a reasonable time (see Sect. 5).

Basic design and policy questions, such as cost estimation in different situations and the service's economies of scale, can be addressed by SOM1, which is the simplest SOM. The same model can also be used to approximate the necessary fleet size and vehicle capacity. SOM2 and SOM3 are used to examine fleet size and vehicle capacity approximations, respectively.

The rest of the paper is organized as follows. Section 2 reviews the relevant background. Section 3 presents a conceptualization framework for DRT feeder service design. Section 4 is dedicated to formulations and solution methods. Section 5 shows the results of our numerical study, and Sect. 6 provides our conclusions and presents opportunities for future research.

2 Background

Innovative transportation services have been studied by numerous researchers under different titles, as can be seen in the recent reviews of semi-flexible transit (Errico et al. 2013); demand-responsive public bus services (Vansteenwegen et al. 2022); and mobility-on-demand (Markov et al. 2021). Some of these ideas were also tested in practice, for example, the EcoBus in Germany (Avermann and Schlüter 2019; Sörensen et al. 2021) and the BRIDJ service in Sydney, Australia (Perera et al. 2020).

The large variety of innovative transportation services creates a challenge when studies are summarized or compared. In this paper, we rely primarily on the classification of transportation services proposed by Markov et al. (2021), which is based on seven key dimensions: access, routing, timing, ridesharing, booking, payment, and pricing. In the access dimension, they differentiate between *coordinate-based* or door-to-door services (such as taxi services), and *station-based* access. Served stations can be *fixed* (as in bus services) or *demand-responsive (which they term "dynamic")*. The routing can be either *fixed-route* (such as bus services), *demand-responsive-route* (such as taxis and various other MoD services), or *mixed-route*. Service timing may be predetermined, denoted as "time-based," as in the conventional PT. Alternatively, service timing can be determined in real time, in response to passenger requests and other dynamic conditions. Such services are denoted as "on-demand." The ridesharing dimension can either be *shared-ride* (or pooled) or *individual-ride* (such as taxi services). Shared-ride services include two subtypes: *professional ridesharing*, which "includes operator-based services, and professional peer-to-peer services such as Uber and Lyft," and *casual peer-to-peer* ridesharing (also known as carpooling). As for booking options, they distinguish between two main options: *reservation-based* services (most on-demand services) and *walk-in* services that do not require reservation (such as a typical bus and metro services). Markov et al. (2021) refer to the latter as "open-access" services. *Reservation-based* services can be further categorized into two subtypes: *prebooking* and *instant-booking* services. On payment and pricing they write: "The payment can be for each *individual* use of the service, for a *package* of a fixed number of uses, or through a *subscription* plan. The subscription plan itself may be limited to the service in question or bundled with other transportation services in the same city or area. Pricing can be *fixed*, as in the case of most time-based services, or *dynamic*."

An additional important dimension for classifying transportation services is their *scope*. MoD studies typically assume that the scope is defined by the covered area (e.g., within a polygon). Other MoD and DRT studies are dedicated to feeder services that "transport passengers from a low-demand area, like a suburban area, to a transportation hub, like a city center" (Montenegro et al. 2020; 2021) or like a train station.

In the literature on general MoD, instant booking is very common. For example, Markov et al. (2021) reviewed 57 MoD studies, of which 47 considered a service with instant booking, and only four considered a service with prebooking. In contrast, among 16 SFLMTS studies we found (Table 2, rows 1–16), prebooking was considered in seven studies, while only four considered a service with instant booking. The six additional SFLMTS studies considered other reservation options, including various combinations of prebooking, instant booking, and walk-in.

Most of the 16 SFLMTS studies assumed on-demand service timing with demand-responsive routing (fully or partially), and they typically assumed professional ridesharing (whether the vehicle is autonomous or manually driven). Nearly

Table 2 Classification of studies on last-mile-transportation-services (SFLMTS)

No	Authors (year)	Access	Booking	Excess time	Stops per route
1	Scheltes and de Almeida Correia (2017)	Coordinate-based	Combination		
2	Wang (2017)	Station-based	Combination		3 to 5
3	Basu et al. (2018)	Coordinate-based	Instant-booking	Yes	
4	Charisis et al. (2018)	Station-based	Prebooking		
5	Raghunathan et al. (2018a)	Station-based	Prebooking		One
6	Raghunathan et al. (2018b)	Station-based	Prebooking		One
7	Shen et al. (2018)	Station-based	Instant-booking	Yes	
8	Sun et al. (2018)	Station-based	Prebooking		
9	Ma et al. (2018)	Coordinate-based	Instant-booking		3 to 5
10	Montenegro et al. (2020)	Station-based	Combination		3 to 5
11	Wang et al. (2020)	Station-based	Prebooking		
12	Leffler et al. (2021)	Station-based	Instant-booking		3 to 5
13	Montenegro et al. (2021)	Station-based	Combination		3 to 5
14	Shehadeh et al. (2021)	Station-based	Prebooking		
15	Wang et al. (2021)	Station-based	Combination		
16	Sangveraphunsiri et al. (2022)	Coordinate-based	Combination		
17	DRT 1-M (this paper)	Station-based (demand-responsive)	Prebooking	Yes	1 to 4
18	FRT 1-M (this paper)	Station-based (fixed)	Walk-in	Yes	1 to 4

all these studies assumed that no passenger rejection is allowed, and most of them included vehicle travel time in the objective function.

Other service attributes differ across studies. For example, most studies assumed station-based service, and four assumed coordinate-based service; yet none of them assumed the combination of coordinate-based booking with station-based service, as we assume here. Most ignored walking time, whether because the service is door-to-door, or because the demand point is at the station. Excess ride time constraints, as assumed in our models, were considered only in two studies (Basu et al. 2018; Shen et al. 2018). Substantial variation can be found in the assumptions about fleet size (2–600) and vehicle capacity (2–50). Most assume that the number of stops along the route is not limited, but several studies considered 3–5 stop routes, and some studies assumed single-stop routes.

Most studies on instant booking used a simulation environment to evaluate the performance of demand-responsive policies such as dispatching rules and other heuristics. Most studies on prebooking focused on a deterministic static setting. They formulated the problem as mixed-integer programs (MIPs) and explored various solution heuristics relying on column generation, tabu search, large-neighborhood-search, genetic algorithms, etc. In addition, all reviewed papers took an operational perspective on the first/last-mile problem, while the perspective in this study focuses on evaluations that can help in the service design process.

One of the goals of the current study is to evaluate the benefits of DRT with prebooking by comparing the resources required for providing similar LOS, using fixed routes service (FRT 1-M) and demand-responsive service (DRT 1-M). To clarify the comparison between these two services and the services explored in the previous studies, rows 17 and 18 in Table 2 present the relevant characteristics of these two services. Additionally, Table 3 describes the main properties of the two services using the typology proposed by Markov et al. (2021). Note that the classification of these services in terms of the timing (or schedule) dimension is not exactly time-based, but it is not on-demand either. Departure time from the train station is synchronized with the train arrival. If trains are punctual, then the departure time is predetermined. Yet, if a train is late, the shuttle will wait for the train passengers. In any case, arrival time to a specific bus stop is route-dependent, and therefore not

Table 3 Primary characteristics of DRT 1-M and FRT 1-M services

Dimension	DRT 1-M	FRT 1-M
Access	Station-based (demand-responsive)	Station-based (fixed)
Routing	Demand-responsive route	Fixed-route
Timing	Time-based (at departure)	Time-based
Booking	Reservation-based: prebooking (coordinate-based)	Walk-in
Ridesharing	Yes. Provided by professional drivers	Same as DRT 1-M
Service level	Limited excess ride time (compared with direct ride) Limited walking time from/to the bus stop Negligible waiting time at the train station No rejections of transportation requests	Same as DRT 1-M

exactly predetermined. However, the time window of arrival is dictated by the constraints on excess ride time. Figure 1 presents the input and output details associated with the DRT 1-M and FRT 1-M optimization module.

To summarize, the main research gap in the literature on DRT 1-M for SFLMTS is primarily that the studies considered an operational rather than design perspectives. As a result, the typical primary question was how to efficiently find an optimal (or near-optimal) routing solution while considering the constraints of a particular service design, which is assumed as given. This paper focuses on how policymakers should address design questions related to SFLMTS, starting from the basic question of whether, in a specific setting, demand-responsive routes provide sufficient benefit compared to fixed routes (which are easier to manage). We then offer tools to evaluate the impacts of LOS and competition, as well as alternatives for cost estimation at the design stage.

3 Service design framework

Our proposed DRT service design framework differentiates high-level aspects from detailed design choices. The high-level aspects we consider are commitment (Sect. 3.1), trip consolidation (Sect. 3.2), and service-level considerations (Sect. 3.3). The separation of the discussion into subsections is a bit artificial, since the various design choices are related to each other. Critical connections are recognized along the way and are more thoroughly addressed in Sect. 3.4, where a summary of the specifications of the services considered in our research is presented.

Part of the design process is comparing alternative variants of the DRT service design presented in Sect. 3.4. These include issues like the value of prebooking, market structure, and LOS parameters. Quantitative evaluation of these alternatives

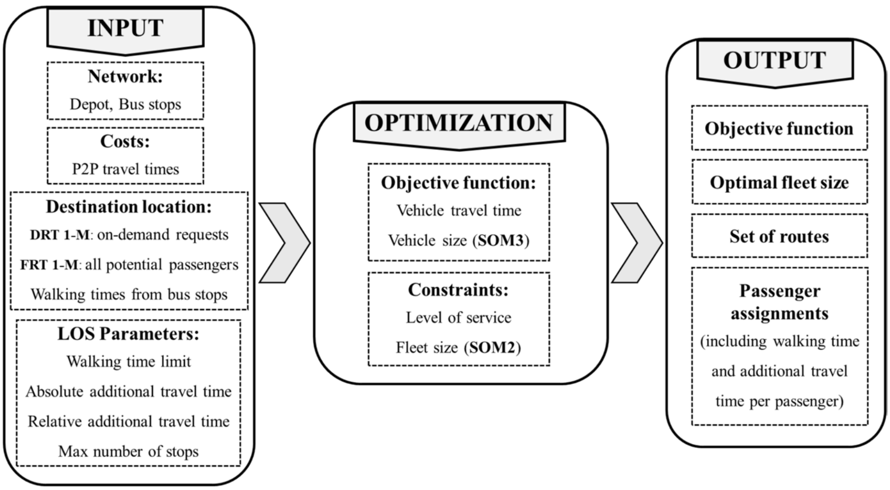


Fig. 1 Optimization module workflow (P2P: point-to-point)

may require the consideration of certain detailed design choices. These include fleet acquisition choices, as well as the mechanism for bus stop choice (Sect. 3.5). Given the design framework of Sects. 3.1–3.5, we present in Sect. 3.6 the structure of our proposed evaluation scheme, which serves as the basis for the analysis in the remainder of the paper.

3.1 Commitment

The commitment of the operator and the passenger is an important aspect in the design of a transportation service, but it is largely overlooked in public transit literature. The key decisions are: What is the operator committed to? What are passengers committed to? And when are these commitments made?

In conventional PT, passengers are not required to commit themselves, while the operator is required to handle the resulting uncertainty. Some operators manage to follow the prescribed schedule and provide sufficient capacity better than others, and in that sense, the level of commitment on the side of the operator varies. Still, there is always some level of commitment. From the passengers' perspective, this is a significant advantage.

In DRT systems, passengers are required to show some level of commitment. In traditional dial-a-ride services, typically, passengers make their reservations the day before, and they are required to be at the pickup point during the pickup time window. The LOS is not guaranteed, which does not cause a serious problem as the passengers are captive and, therefore, competitiveness is not an issue.

Instant booking DRT systems (e.g., e-hailing) require passengers to commit themselves by accepting the offer, and to be available for pickup from the moment of the reservation and onwards. Some of these services guarantee a fixed price and a limited waiting time at the time of reservation (e.g., Uber-pool[1] or Via[2]), while others do not make specific guarantees. A key limitation of many DRT services is that service availability is not guaranteed, and thus, requests may be rejected at the discretion of the operator.

Train service is typically scheduled, and often with relatively long gaps between departures. Therefore, passengers may expect a reliable feeder service, and an operator may be required to commit to service availability for all requests. Ensuring an inclusive service of this type is not trivial at all. A key question in our research is the viability of such a service.

Passenger commitment in the train alighting scenario can be made by passengers once they are on board the train. Passengers can place their order using a smartphone application, and they will have to specify the train station they are headed to, the estimated train arrival time, and their final destination address. Assuming that the train ride is at least 15 min long, the operator gets a decent opportunity to process all requests, explore alternative assignments of passengers to vehicles, and

[1] https://www.uber.com/au/en/ride/uberpool/.

[2] https://ridewithvia.com/.

inform all passengers about the specific vehicle they should take from their alighting train station to their destination.

In the train boarding scenario, it may be reasonable to require passenger commitment even further in advance, say 1 h or so before the planned train departure, since additional time is needed for the actual travel to the train station and for the transfer from the feeder service to the train. Additional time could also be needed if passengers are requested to walk from their origin to a nearby pickup location. In both contexts of alighting or boarding a train, passengers are likely to know their exact plan when they make the commitment, and, therefore, the added burden by the required commitment as described above appears relatively modest.

Note that in other DRT niches, for example at employment centers, temporal commitments are not necessarily inherent to existing travel patterns and may thus impose a non-negligible additional burden. The question of whether this is indeed the case, and if so, how big the impact on utilization is, remains a subject for future research.

3.2 Trip consolidation

An important common feature of PT and DRT transit systems is their nondiscretionary ride-sharing nature, in contrast to programs where ride sharing is at the passengers' discretion, like carpooling in dedicated high-occupancy-vehicle (HOV) lanes. Trip consolidation mechanisms are, therefore, critical for increasing occupancy and achieving efficiency.

Trip consolidation involves temporal and spatial dimensions (Tachet et al. 2017). Consolidation can be partly spontaneous and partly induced. Spontaneous consolidation happens when passengers have a similar (or identical) desired origin, destination, departure time, and arrival time. During peak hours, especially in a dense urban area, like a central business district (CBD) of a large metropolitan area, most of the consolidation is spontaneous. Under such circumstances, offering shared transport services, PT or DRT, is much easier.

Induced consolidation means utilization of passengers' flexibility regarding one or more of their trip attributes, i.e., passengers adjust their trip attributes to enable trip consolidation. There are several ways to induce trip consolidation. In PT, bus stops induce spatial consolidation, while predetermined schedules induce temporal consolidation. Thus, passengers' flexibility is utilized to achieve efficiency while enabling operators to provide commitment. A location adjustment consolidation mechanism, such as walking to a bus stop or to the nearest corner, is also used by some DRT systems, like Via. Departure time adjustment as a consolidation mechanism is used by some pre-ordered DRT systems, like dial-a-ride.

When DRT is used as a feeder service for trains, e.g., in the train alighting scenario, despite a wide spread of the destinations, trip consolidation can still be relatively easy, as the train station itself is the origin of all feeder trips and can, therefore, operate as the depot of the service. Departure times are also identical to all passengers alighting from the same train, thus temporal consolidation is solved as well. In principle, the departure time is known in advance, but even if the train service is delayed, all passengers on board the same train incur the same delay, and, therefore, the possibility for trip consolidation is not affected. Further spatial consolidation can be achieved if several passengers are dropped off at one common bus stop, within reasonable walking distances from their respective destinations. In terms of trip consolidation, the one-to-many service for train alighting is completely symmetric to the many-to-one service for train boarding.

Many-to-one and one-to-many DRT services can be considered in other niches, by placing the depot where activities are dense, e.g., at airports, employment centers, university campuses, community centers, etc. In terms of the spatial context of trip consolidation, these niches are similar to the train feeder service context. To enable temporal consolidation, the operator may need to impose a fixed schedule, e.g., depot arrival/departure every 15 min during the peak hours. Discussion and evaluation of such additional temporal consolidation schemes are beyond our current scope.

3.3 Service-level considerations

Earlier implementations of DRT, i.e., dial-a-ride, were designed to serve captive passengers. As such, competition was not a major concern. In recent years, the focus shifted to DRT services that must compete, at least with PT and taxis, and possibly with private car driving as well. Passenger choices depend on LOS and price. Primary LOS attributes are walking time to or from the pickup and drop-off locations and excess ride time. Other components of the LOS that are relevant to some DRT systems are the availability of the service and response time. However, these are irrelevant to the time-based service studied here.

The walking time and excess time of DRT are inferior to taxi, almost "by definition." To be viable, the LOS should be as close as possible to taxi, and much better than conventional PT, while the price should be as close as possible to PT, and much lower than taxi. Furthermore, if the service is prebooked (see Sects. 3.1 and 3.2), both price and LOS must be guaranteed by the operator upfront. This implies that the operator should consider LOS as a constraint.

The constraint on ride time could be stated by an additive constant delay compared to direct taxi service, for example, up to five additional minutes. It could be stated in relative terms, for example, 20% of excess ride time. It could also be determined by passenger priority, distinguishing premium passengers from others. Walking time constraints can be based on a fixed limit for all passengers or consider passenger classifications as well.

It should be noted that there are other important aspects of LOS relevant to the studied system that are not captured by the model presented in this paper; e.g., the type of vehicles and how crowded they are, service punctuality, physical conditions in the stations and on the vehicles, etc.

3.4 Service specification summary

The service we focus on is the one-to-many DRT (DRT 1-M), where a given batch of train passengers, who share their alighting train station, and therefore their alighting time, travel to different final destinations. The feeder service we consider is an on-demand shuttle service from the train station to bus stops close to the respective destinations of all the passengers in the batch (passengers of a specific train). Passengers commit to using the service by prebooking, while providing information on their desired destination. At some cut-off time before the arrival of the train, the operator processes all requests in the batch, selects routes for the vehicles, and assigns the passengers to them.

Prior to departure, the fleet is in the terminal, located at the train station, and the vehicles must return to the terminal after dropping off all the passengers. Since the departure times of the feeder service are adjusted to the train arrival times, the number of trips that will serve customers of a specific train is equal to the number of vehicles required to complete the service.

Like some previous studies, we assume that the operator is allowed to drop off passengers only at predefined locations, such as bus stops in the service area. Czioska et al. (2019) refer to these locations as "meeting points." They explain that pre-selected locations may be advantageous in terms of stopping convenience and passenger safety and show numerical examples where serving the same set of passengers at meeting points reduces the total distance that vehicles travel by up to 30%.

Note that in the service we consider, several passengers whose destinations are different may be dropped off at the same bus stop. After the assignment, each passenger is informed about the specific vehicle they should board.

The operator is committed to respond to all requests, while guaranteeing a predetermined LOS in terms of maximum allowed walking time of each passenger and maximum allowed excess ride time (destination-dependent), as well as the number of stops on the route. Allowed excess ride time is guaranteed relative to a direct trip to the drop-off bus stop by a special taxi or a private car and includes two components: 1. relative excess ride time, compared to the direct travel time; 2. additional constant delay.

3.5 Detailed design choices

Implementing a DRT service based on the specifications presented in Sect. 3.4 requires various additional design choices. These may be determined by the regulator, by the operator, or jointly. We focus here on two of these choices that may influence the evaluations of alternatives, as discussed in Sect. 3.6. The first issue is fleet acquisition choices, and the second is bus stop choices.

Fleet acquisition involves choices about fleet size, vehicle capacity, and driver availability. These are major cost components that should be considered, at least approximately, during the design process. The challenge is the gap between the design and the operation stages. Fleet acquisition operation can be managed in various ways. In operational models, it is common to assume that fleet choices are fixed, rather than demand-responsive, but operators may have a certain level of flexibility even after passengers submit their requests. At the design stage, it may be sensible to assume that the fleet is determined according to an extreme scenario (highest demand), which may imply that in most cases, fleet constraints are not binding.

The second issue is bus stop choice. To explain the issue, consider, for example, a batch served by three shuttles (A, B, C), and suppose that the destination of a specific passenger is within the walking time limit from three stations: the second stop of the first shuttle (A2), the third stop of the first shuttle (A3), and the first stop of the third shuttle (C1). The operator may present all three options to the passenger, allowing them to choose. The operator may choose one option for the passenger, based on predefined LOS criteria, like shortest walking time or earliest arrival time. The criteria may be the same for all passengers, or could be selected by each passenger, using an app-setting menu. Finally, the operator may choose (or restrict) shuttle assignment based on vehicle capacity considerations. In the latter case, an operator may be interested in exploring the Pareto front of vehicle capacity against other cost components (e.g., total vehicle travel time). If operator priorities are known, these could be considered in the design process. Otherwise, it may be sufficient for design purposes to consider one Pareto solution, such as the solution that minimizes the maximum vehicle occupancy, without an increase to other cost components. This option is addressed in our research by SOM3.

3.6 Quantitative evaluation scheme

There are several variants of the above-mentioned DRT 1-M service that may be considered during the design stage. The value of demand-responsive routes compared with fixed-route-transit (FRT) may be questioned. The effect of market structure and the number of competing operators from the same train station could be of interest. Clearly, the effect of LOS parameters on operating costs is also important during the service design stage.

Addressing these questions requires a quantitative evaluation scheme. As discussed in the introduction, we propose to consider three SOMs for this purpose. All three SOMs assume that the operator minimizes total vehicle time traveled, subject to constraints on LOS and inclusivity (requests cannot be rejected). Train arrival frequency is assumed to be low, thus justifying the consideration of each train arrival separately.

Most of the evaluations presented here are based on SOM1, where fleet size and vehicle capacity are assumed to be non-binding (and therefore unconstrained in the optimization model). The outputs of SOM1 can be used as estimates for the desired fleet size and vehicle capacity, and thus provide a basis for operating cost estimates. SOM2 adds a fleet size constraint for examining the trade-off (if exists) between fixed costs (fleet size) and variable costs (total vehicle travel time). SOM3 adds a secondary objective of minimizing the maximal occupancy, thus providing a tighter estimate of the necessary vehicle capacity. Such estimates are often sufficient for the design stage.[3]

The proposed SOMs focus on the needs of the design stage, and deliberately ignore various operational complexities. For example, the contract between the regulator and the operator should consider the possibility that occasionally (hopefully rarely) deviations from the commitment for LOS and inclusivity may occur. Similarly, policies should be defined for passenger cancelations. These issues may be part of the detailed service design, but they are not relevant for the preliminary design process we discuss here.

As part of the qualitative evaluation, we also consider a reference service where each passenger is served by a different vehicle. To provide a meaningful comparison, we assume that these vehicles will drop off their passengers at the closest bus stop to their destination. We refer to this reference service as taxi-to-bus-stop (TBS).

A concluding observation regarding the pros and cons of simplified optimization models is warranted. As the model is streamlined, achieving optimal solutions generally becomes more straightforward, offering a significant advantage when assessing different alternatives during the service design phase. Service operation should consider cancelations, uncertainties, mode choice, differences between commuters and occasional travelers, multiple train arrivals during short intervals, impacts of the specific time of day (e.g., peak or off-peak), etc. These operational considerations are not included in the models we discuss here.

4 Modeling and algorithm

In this section, we present the DRT 1-M problem that can be verbally defined as follows. Given a set of passengers initially located at the depot, where each passenger is characterized by a set of possible drop-off bus stops, find a set of cyclic routes that start and end at the depot and visit at least one bus stop from the desired set of each passenger. The set of possible drop-off bus stops for each passenger is determined in a preprocessing step and therefore considered in the mathematical models as given. Each route may visit up to a predefined number of bus stops. Any visited bus stop must be visited before a given (stop-specific) time, which we refer to as the closing of its time window. The objective is to minimize the total length (travel time) of all the routes. Equivalently, the same model is applicable to the design of a feeder service based on fixed lines by taking as input a set of potential passengers rather than a set of actual passengers in a specific batch.

Because the vehicles in this model are not subject to capacity constraints and the travel times satisfy the triangle inequality, the routes in an optimal solution are disjoint, which is often referred to as “no split deliveries” property.

For our purposes, the set of bus stops associated with each passenger represents locations within a reasonable walking time from their final destination (say 6 min), and the closing of the time window at each bus stop represents a service level

[3] If an evaluation of the trade-off between vehicle capacity and variable costs is needed during the design stage, alternative models may be needed, which remain a subject for future research.

constraint that is calculated relative to the direct travel time from the depot to the bus stop location.

A mathematically equivalent problem is the DRT M-1 problem when many passengers need to be picked up from bus stops near their destination subject to time window opening constraints. In this study, the presentation and analysis are in terms of the 1-M (outbound) version of the problem, but the results are also valid for the M-1 (inbound) version.

We start with an explicit arc-indexed formulation of the problem. Since it is a variant of the VRP with time windows, our model is inspired by a well-known formulation from the literature (see, for example, Toth and Vigo, 2014, page 121, Eqs. 5.1–5.9) with the following modifications:

1. Instead of visiting all the customers, the routes should cover a subset of the bus stops that are determined endogenously by the model.
2. The closings of the time windows are determined by the service-level constraints based on the direct travel time from the depot to each bus stop. All the time windows open at time zero.
3. The number of stops per route is limited. This is similar to a distance-limit VRP (DVRP).

We solve this problem by the set covering approach, using complete route enumeration. Our results show that practical instances of our problem can be solved to optimality in reasonable computation times by this approach. There are several reasons for the viability of the complete enumeration approach in our case, compared to the previous studies:

1. Tight service-level constraints limit the number of feasible routes substantially compared with other variants of the VRP.
2. Advancements in algorithms dedicated to the set covering problem enable practical solutions using a generic solver package.
3. The same set of routes applies to multiple demand instances, and, therefore, the route-enumeration effort is done only once and can be reused for many scenarios as long as the set of bus stops is not changed.
4. The vehicle capacity constraint is omitted as it is less important in the service design stage.

Since complete route enumeration is not a common practice in VRP studies, we provide a pseudocode for the efficient generation of all feasible non-dominated routes using a dynamic programming approach. We then explain how the method can be used in our specific case.

The remainder of the section is organized as follows. The mixed-integer linear formulation of the problem is presented in Sect. 4.1. Section 4.2 proposes an alternative set covering formulation, as well as an efficient dynamic programming algorithm for route generation. The issue of passenger assignment, especially in SOM3, is addressed in Sect. 4.3.

4.1 Arc-indexed formulation

Here, we present first the arc-indexed mixed-integer linear program (MILP) for the DRT 1-M problem. The purpose of this formulation is primarily to provide a tight description of the problem we want to solve.

We divide the problem parameters into three categories: network attributes, service level parameters, and demand realizations. In the numerical results section, we consider a single network, several alternatives regarding the service-level parameters, and multiple demand realizations. The network attributes are:

S Set of all bus stops in the service area $(1, \dots, n)$, excluding the depot.

S_0 Set of all possible stops, including the depot (denoted by 0), i.e., $S_0 = S \cup \{0\}$.

A Set of arcs in the underlying complete directed graph, i.e., $A = \{(i,j) : i,j \in S_0, i \neq j\}$.

c_{ij} Non-negative travel cost for each arc $(i,j) \in A$, which represents the vehicle travel time from stop i to stop j. The travel times satisfy the *triangle inequality*, i.e., $c_{ij} + c_{jk} \geq c_{ik}$ for all $i, j, k \in S_0$.

V The number of vehicles in the fleet (if relevant as in SOM2 defined above).

The service level parameters are:

K Maximal number of bus stops along each vehicle route (not including the depot).

α Allowed relative excess travel time of the vehicle compared to the direct travel time to a drop-off bus stop. Presumably $\alpha \geq 1$.

β Allowed additive excess travel time of the vehicle compared to the direct travel time to a drop-off bus stop. Presumably $\beta \geq 0$.

φ_i^{max} Maximal allowed travel time of a vehicle to bus stop i, $\varphi_i^{\max} = \alpha \cdot c_{0i} + \beta \geq c_{0i}$.

A demand realization is described by:

P A batch (set) of actual (or potential) passengers. Each passenger has an associated destination.

w_{ip} Walking time from drop-off bus stop $i \in S$ to the destination of passenger $p \in P$.

w_p^{max} Maximum walking time of passenger $p \in P$ from the drop-off bus stop to his destination (can be the same for all passengers).

The original inputs are preprocessed to obtain the following auxiliary inputs:

$S(p)$ Set of permissible bus stops at which passenger $p \in P$ can be dropped off, $S(p) = \left\{ i \in S_0 : w_{ip} \leq w_p^{max} \right\}$. Note that by the construction of this set, we can limit the walking perimeter of the passengers, possibly according to different criteria for each one of them.

S^P Set of all possible drop-off bus stops, which are relevant for the given batch of passengers

$$S^P = \left\{ \bigcup_{p \in P} S(p) \right\} \subseteq S_0.$$

S_0^P Set of all batch-relevant stops in the problem, including all possible drop-off bus stops, as well as the depot, i.e., the pickup location at the train station, $S_0^P = S^P \cup \{0\}$.

The decision variables in our model are:

x_{ij} Binary variable that equals "1" if a vehicle travels directly from stop i to stop j.

z_{ip} Binary variable that equals "1" if passenger p is dropped off at bus stop i, defined for all pairs of *(i, p)* such that $i \in S(p)$.

φ_i Arrival time of a vehicle at stop i (where $\varphi_O = 0$).

ψ_i Number of stops on the route to stop i (where $\psi_O = 0$).

The mathematical formulation of DRT 1-M is

$$min \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{1a}$$

$$\sum_{i\in S_0^P} x_{ij} \le 1 \qquad \forall j \in S^P \tag{1b}$$

$$\sum_{j\in S_0^P} x_{ij} = \sum_{k\in S_0^P} x_{ki} \qquad \forall i \in S_0^P \tag{1c}$$

$$z_{ip} \le \sum_{k\in S_0^P} x_{ki} \qquad \forall p \in P, i \in S(p) \tag{1d}$$

$$\sum_{i\in S(p)} z_{ip} = 1 \qquad \forall p \in P \tag{1e}$$

$$\varphi_j \ge \varphi_i + c_{ij} - \left(\varphi_i^{max} + c_{ij}\right)\left(1 - x_{ij}\right) \qquad \forall i \in S_0^P, j \in S^P \tag{1f}$$

$$\varphi_i \le \varphi_i^{max} \qquad \forall i \in S^P \tag{1g}$$

$$\psi_j \ge \psi_i + x_{ij} - \left(1 - x_{ij}\right)K \qquad \forall i \in S_0^P, j \in S^P \tag{1f}$$

$$\psi_i \le K \qquad \forall i \in S^P \tag{1i}$$

$$\sum_{i\in S^P} x_{o,i} \le V \tag{1j}$$

$$x_{ij} \in \{0,1\} \qquad \forall (i,j) \in A \tag{1k}$$

$$z_{ip} \in \{0,1\} \qquad \forall p \in P, i \in S(p) \tag{1l}$$

$$\varphi_i, \psi_i \ge 0 \qquad \forall i \in S_0^P. \tag{1m}$$

The objective function (1a) minimizes the total travel times of the vehicles. Constraint (1b) requires that at most one vehicle arrives at each bus stop, since there are no "split deliveries" in an optimal solution. The flow conservation Eq. (1c) ensures that a vehicle exits a bus stop if and only if a vehicle enters that bus stop. Constraint (1d) stipulates that if a passenger is dropped off at a bus stop, a vehicle must arrive there. Constraint (1e) stipulates that each passenger is dropped off at exactly one permissible bus stop. Compliance with the maximum walking time allowed for passenger p is guaranteed by the definition of the set of permissible drop-off bus stops $S(p)$. Constraint (1f) assigns the arrival time of a vehicle to bus stop j to the φ_j variable. We use the fact that the arrival time at each bus stop i cannot be larger than φ_i^{max}. Thus, if the arc (i, j) is not traversed (i.e., $x_{ij} = 0$), then (1f) always holds. Constraint (1g) assures that the ride time of each passenger p does not exceed the maximum allowed time φ_i^{max}. Constraint (1h) counts the number of bus stops of a vehicle per route at each bus stop j and ensures that it is incremented by one, if the arc (i,j) is traversed (i.e., $x_{ij}=1$). Otherwise, (i.e., $x_{ij}=0$) (1h) always holds. Constraint (1i) assures that the number of bus stops of a vehicle per route is not greater than the allowed maximum number of bus stops. Constraints (1f) and (1h) also ensure subtour elimination, following the well-known idea of Miller, Tucker, and Zemlin (1960). Constraint (1j) stipulates that at most V vehicles are used, which is used as an active constraint in SOM2, and relaxed in SOM1.

The MILP formulation ($1a$-$1l$) is presented here to provide a coherent exposition of the DRT 1-M problem, in a compatible form with other VRP formulations. Even though this formulation is relatively simple, our attempts to use it with a commercial solver yielded relatively poor results, for instances with more than 20 passengers and about 70 potential drop-off locations. Adding some valid inequalities improved the results, but only slightly. In the next section, Sect. 4.2, we present an alternative formulation that allows solving much larger instances using commercial solvers at a reasonable time.

4.2 Set covering formulation and route enumeration

Many exact and heuristic solution approaches to VRPs divide the problem into two subproblems, namely a route generation problem and a set covering problem. Our implementation of this approach consists of three stages: offline preparation of a batch-agnostic route collection, screening a batch-specific route collection, and identifying an optimal solution by solving a set covering problem.

A closed depot-based route is a cyclic sequence of bus stops that begins and ends at the depot. A route is feasible if its arrival time at each stop i is not greater than the maximum allowed travel time at this stop, φ_i^{max}. A feasible route is *non-dominated* if it is one of the shortest routes that visit the same set of stops.

As a preprocessing step, we construct a batch-agnostic route collection $\mathcal{R}$ as a set of all the depot-based feasible non-dominated routes (excluding ties) that visit up to K bus stops. For each route $r \in \mathcal{R}$, let B_r denote its set of bus stops and c_r be its total travel time, including the arc from the last stop back to the depot. We note that in case of a tie, i.e., $B_{r_1} = B_{r_2}$ and $c_{r_1} = c_{r_2}$, only one of two routes should be included in $\mathcal{R}$.

Once the demand (of a specific batch P) is revealed, we select from $\mathcal{R}$ a subset $\mathcal{R}^P = \{r \in \mathcal{R} : B_r \subseteq S^P\}$. Let $P_r = \{p \in P : B_r \cap S(p) \neq \emptyset\}$ be the set of all passengers that can be served by each route $r \in \mathcal{R}^P$. An optimal solution of DRT 1-M can be constructed by selecting routes from $\mathcal{R}^P$ that jointly cover all the passengers in P with minimum total travel time. This set covering problem can be formulated as the following integer programming model:

$$\min \sum_{r \in \mathcal{R}^P} c_r \cdot x_r \tag{2a}$$

Subject to:

$$\sum_{r \in \mathcal{R}^P : p \in P_r} x_r \geq 1 \qquad \forall p \in \mathrm{P} \tag{2b}$$

$$\sum_{r \in \mathcal{R}^P} x_r \leq V \tag{2c}$$

$$x_r \in \{0,1\} \qquad \forall r \in \mathcal{R}^P. \tag{2d}$$

The objective function (2a) minimizes the cost of all the selected routes, and constraint (2b) stipulates that all the passengers are covered and (2c) that at most V vehicles are used. We claim that the arc index formulation (1a)-(1m) is equivalent to the set covering formulation (2a)–(2d) if $\mathcal{R}$ is constructed correctly, as described below.

Next, we present a dynamic programming (DP) algorithm inspired by the classical DP approach for the traveling salesman problem (TSP) to enumerate the set of non-dominated routes $\mathcal{R}$. The process is divided into two steps: (1) generating collections of all DP states, each associated with a feasible non-dominated open route (defined below), and computing its cost; (2) Using the set of open non-dominated routes to create the set of non-dominated closed routes, $\mathcal{R}$.

A feasible open route is a sequence of stops that starts at the depot and respects the maximum allowed travel time constraint at each stop. It is said to be non-dominated if it is not longer than any other feasible open route that visits the same set of stops and ends at the same last bus stop. Each non-dominated open route is associated with a state (B, i), such that $B \subseteq S$ is the set of bus stops visited by it, and $i \in B \cup \{0\}$ the last stop in the sequence. Note that if B is not empty, then the position of the vehicle at the end of the open route must be one of its bus stops, and hence, $i \in B$. The state $(\emptyset, 0)$ represents the empty open route that does not visit any bus stop. We point out that B is an unordered set, i.e., the state does not explicitly specify the sequence of the stops in the associated route.

The collection of states of length k, i.e., $|B| = k$, is denoted by Ω_k, and $\Omega_0 = \{(\emptyset, 0)\}$. For a state, (B, i), we denote the state cost by $C(B, i)$, representing the total cost (vehicle travel time) of a non-dominated open route that starts from the depot, ends at i and visits all the bus stops in B. By definition, $C(\emptyset, 0) = 0$. A forward Bellman equation for state costs can be defined as follows:

$$C(B \cup \{i\}, i) = \min_{j \in B; (B,j) \in \Omega_k} \{C(B, j) + c_{ji}\} \qquad \forall B \subseteq S, |B| = k, 0 \leq k < K, i \in S \setminus B. \tag{3}$$

Let $\Omega = \{B \subset S | \exists i \in B, (B, i) \in \Omega_{|B|}\}$ be the collection of all the feasible sets of bus stops that can be served in a single route (open and closed). For every feasible set of bus stops, $B \in \Omega$, we can find the minimal cost of a closed feasible route that begins at the depot, serves the bus stops in B, and returns to the depot, by selecting the last bus stop i that minimizes the cost after adding the cost of deadheading the vehicle back to the depot. That is

$$C_0(B) = \min_{i \in B}\{C(B, i) + c_{i0}\} \qquad \forall B \in \Omega. \tag{4}$$

Algorithm 1 describes the first step, where the collections of states (Ω_k) are identified, and the state costs are calculated. In the second step (Algorithm 2), we find the collection of all feasible sets of bus stops, Ω, and their costs. We then generate the route collection $\mathcal{R}$, such that for every $B \in \Omega$ there is a feasible, minimal-cost, closed route $r \in \mathcal{R}$.

Algorithm 1 (step 1).

Enumeration of all non-dominated open routes and their cost

Input: network attributes (S_0, c_{ij}); route feasibility parameters (K, φ_i^{max})

Output: a collection of feasible states (Ω_k $\forall\, 0 \le k \le K$); state costs $C(B,i)$ and trace $j^*(B,i)$.

1 $\Omega_0 \leftarrow \{(\emptyset, 0)\}$, $c(\emptyset, 0) \leftarrow 0$;
2 for all $k = 1\ to\ K$ do
3     $\Omega_k = \emptyset$
4   for $(B, j) \in \Omega_{k-1}$ do
5   for $i \in S \setminus B$ do
6   if $C(B,j) + c_{ji} \le \varphi_i^{max}$
7   If $(\mathrm{B} \cup \{i\}, i) \notin \Omega_k$
8                     $\Omega_k \leftarrow \Omega_k \cup (\mathrm{B} \cup \{i\})$
9                     $C(\mathrm{B} \cup \{i\}, i) \leftarrow C(B,j) + c_{ji}$
10                     $j^*(\mathrm{B} \cup \{i\}, i) \leftarrow j$
11   elseif $C(B,j) + c_{ji} < C(\mathrm{B} \cup \{i\}, i)$
12                     $C(\mathrm{B} \cup \{i\}, i) \leftarrow C(B,j) + c_{ji}$
13                     $j^*(\mathrm{B} \cup \{i\}, i) \leftarrow j$

In line 1 of Algorithm 1, the state collection Ω_0 is constructed, containing the empty route that visits an empty set of bus stops, $B = \emptyset$. The value of this state is 0 ($C(\emptyset, 0) = 0$), because no costs have yet been accumulated. In lines 2–13, the algorithm loops through the possible number of stops, k, from 1 to K. In line 3, the state collection Ω_k is initialized to be empty. In lines 4–13, the algorithm loops through all the states of Ω_{k-1}, i.e., the states with k-1 bus stops. In lines 5–13, for each state, the algorithm scans all the feasible extensions by one additional bus stop that is not served yet. In line 6, we check if the bus stop is a feasible extension of the open route represented by the previous state from Ω_{k-1}. If the route obtained by the extension has not been identified yet (line 7), it will be added to Ω_k (line 8), its cost will be calculated and stored with the state (lines 8), and its ancestor state will be traced (line 9). If it is not a new state (else), in lines 11–13, we check if the new route dominates the previous one encountered for the state, and if necessary, update its cost and trace.

Next, we use the states generated by Algorithm 1, each encoding a non-dominated open route, to generate a set of non-dominated closed routes. This is achieved by applying (4) and using the trace variables stored with the states.

Algorithm 2 (step 2).
Enumeration of all non-dominated closed routes

Input: network attributes (S_0, c_{ij}); route feasibility parameters (K, φ_i^{max}); The output of Algorithm 1, collection of feasible states (Ω_k); state costs $C(B,i)$ and traces $j^*(B,i)$.

Output: a collection of all non-dominated closed routes $\mathcal{R}$ and their costs c_r.

$\Omega = \emptyset$
for all $k = 1\ to\ K$ do
  for $(B,j) \in \Omega_k$ do
    if $B \notin \Omega$
      $\Omega \leftarrow \Omega \cup \{B\}$
      $C_0(B) \leftarrow C(B,j) + c_{j0}$
      $j_0^*(B) \leftarrow j$
    else
      if $C(B,j) + c_{j0} < C_0(B)$
        $C_0(B) \leftarrow c(B,j) + c_{j0}$
        $j_0^*(B) \leftarrow j$
$\mathcal{R} \leftarrow \emptyset$
for all $B \in \Omega$ do
  add a route index r to $\mathcal{R}$ with the following initial attributes:
  $B_r \leftarrow B; c_r \leftarrow C_0(B)$
  $B^{'} \leftarrow B$
  $i \leftarrow j_0^*(B);\ \ S_r = empty\ sequence$
  While $B' \neq \emptyset$ do
    $S_r.insert(i)$
    $B^{'} \leftarrow B^{'} \setminus \{i\}$
    $i \leftarrow j^*(B',i)$

Algorithm 2 is divided into two phases. First, it generates the collection of the sets of bus stops in non-dominated routes, Ω. Then, it reconstructs the actual sequences in these routes. In Line 1, Ω is initialized to be empty. In lines 2–11, the algorithm loops through the collections of states $\Omega_1, \ldots, \Omega_K$. In lines 3–11, the algorithm loops through all the states (B,j) from Ω_k, and finds the optimal closed route for each feasible set of bus stops B. For each feasible set of bus stops, B, the algorithm stores the last bus stop (before the train station) in $j_0^*(B)$. The trace variables $j^*(B, i)$ and $j_0^*(B)$ are then used in lines 12–21 to reconstruct an optimal route for each set of bus stops. For each route, the algorithm creates a sequence, S_r, of the bus stops. We use the operator *S.insert(i)* to add an element i at the beginning of sequence S.

The correctness of the proposed solution method is proved in Appendix A.

4.3 Passenger assignment flexibility implications

The DRT 1-M model presented in the previous sections assumes unlimited vehicle capacity and does not determine the passengers' assignment to vehicles. Instead, it guarantees coverage of bus stops sufficiently close to all the passengers.

As discussed in Sect. 3.5, once vehicle routes are given, the operator may deal with passenger assignment to vehicles in several different ways. The operator may leave some flexibility to passengers and let them choose among relevant options according to their own preferences (e.g., walking time, arrival time, etc.). Alternatively, the operator may specify a unique vehicle assignment for each passenger and thus address vehicle capacity limitations, at least in some cases. The possibility of assigning passengers to vehicles may affect decisions regarding the composition of the fleet and thus may be of interest also for design purposes.

In the current section, we present a method for quantifying the level of flexibility that exists within optimal solutions regarding total travel time in terms of passenger assignment to vehicles and its potential implications on vehicle capacity needs and on passenger LOS measures (walking time and arrival time).

Assignment flexibility occurs, since the optimal vehicle routes may enable some passengers a choice between multiple optional drop-off stops along a specific route (referred to as "within-route flexibility"). In addition, passengers may have a choice between several vehicle routes (referred to as "between-routes flexibility").

Flexibility between routes affects vehicle occupancy and is, therefore, more important than within-route flexibility. The optimal solution, x_r^*, of (2a-2d) for demand realization P can be represented by a set $\mathcal{R}^{P*} = \left\{ r \in \mathcal{R}^{\mathrm{P}} : x_r^* = 1 \right\}$). To quantify the between-routes flexibility, we define a binary matrix C^P (of size $|P| \times \left|\mathcal{R}^{P*}\right|$) that represents the passenger coverage by the routes found in the optimal solution, where C_{pr}^P is one if passenger p can be served by route r, and zero otherwise.

For example, assume that when booking a service for four passengers, the optimal solution found three routes (A, B, C) with the following possible coverage:

$$C^P = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

The first passenger can choose route A or route C, illustrating between routes flexibility. If the passenger can choose two alternative stops along route A, e.g., A2 or A3, this is considered as within route flexibility, which is not shown in the coverage matrix.

Let $H_p^P = \sum_{r \in \mathcal{R}^{P*}} C_{pr}^P$ denote the number of trip alternatives for passenger $p \in P$ in the optimal solution (for example, in the given example $H_1^P = 2$ and $H_4^P = 1$). We define the sum of trip alternatives for all passengers for demand realization P by $H^P = \sum_{p \in P} H_p^P = \sum_{p \in P} \sum_{r \in \mathcal{R}^{P*}} C_{pr}^P$. The average flexibility for a specific demand realization is $F^P = \left(\frac{H^P}{|P|} - 1 \right) \cdot 100\%$. In the example given above, there are five trip alternatives for four passengers, i.e., $H^P = 5$ and $F^P = 25\%$. Finally, given a specific batch size and route length, we define the *flexibility rate* considering a sample $\mathcal{I}$ of demand realizations as follows:

$$Flexibility\,rate = \frac{\sum_{P \in \mathcal{I}} F^P}{|\mathcal{I}|} \tag{5}$$

To examine the implications of passenger assignment flexibility, we propose to compare the solutions of SOM1, where passengers get maximal flexibility, and SOM3, where the operator assigns passengers to vehicles in a way that minimizes the maximum occupancy of all vehicles.

Let y_{rp} indicate whether a passenger $p \in P$ will be assigned ($y_{rp} = 1$) or not assigned ($y_{rp} = 0$) to a route $r \in \mathcal{R}^{P*} : p \in P_r$. Additionally, let Q^{max} be the maximum occupancy of vehicles in the fleet. SOM3 consists of solving the same DRT 1-M problem as SOM1, and then solving the following additional passenger assignment problem (PAP):

$$min\ Q^{max} \tag{6a}$$

Subject to:

$$\sum_{r \in \mathcal{R}^{P*} : p \in P_r} y_{rp} = 1 \qquad \forall p \in P \tag{6b}$$

$$\sum_{p \in P_r} y_{rp} \leq Q^{max} \qquad \forall r \in \mathcal{R}^{P*} \tag{6c}$$

$$y_{rp} \in \{0, 1\} \qquad \forall r \in \mathcal{R}^{P*}, p \in P \tag{6d}$$

The set of constraints (6b) ensures that each passenger is assigned to exactly one of the selected trips. The second set (6c) ensures that the total number of passengers to be assigned to each trip does not exceed the decision variable Q^{max}, which is minimized by the objective function (6a). Note that for a given value of Q^{max}, the problem (6) is in fact finding an optimal solution to a transportation problem. Therefore, it is easily solvable.

5 Computational experiments

To demonstrate the viability of the DRT 1-M feeder service to regional train stations, we applied our proposed evaluation method to a realistic case study and performed a sensitivity analysis. We constructed the case study input from information collected in a survey conducted by Adalya Ltd. on behalf of the Israeli Ministry of Transportation at various train stations in Israel during 2011–2014. These surveys include information about the desired destination address (city and street) of each participant, thus providing a realistic view of the geographic distribution of train passenger addresses. As a case study, we focus on the train station of Carmiel, a town in the north of Israel with a population of about 46,000 (as of 2022). 288 passengers from Carmiel answered Adalya's survey and provided their address details. The train station is the end terminal of the lines that arrive at Carmiel from the city of Haifa 25 times a day. There are 178 bus stops in the town, according to the GTFS file downloaded from the website of the Ministry of Transportation. We obtained walking times and free-flow vehicle travel times from the open street map routing module (OSRM). We assumed a pickup time of 15 s per stop.

For personal taxi-to-bus-stop (TBS), both operational costs and LOS are determined by c_{i0}, the direct vehicle travel time from Carmiel train station to bus stop i (i = 1, 2, …, 178), selected to minimize walking time to the destination. For DRT and FRT, the required LOS in terms of the maximum allowed walking and ride times was, respectively, in minutes, 6 and $3 + 1.3 \bullet c_{i0}$, meaning that the maximum walking time is 6 min and the maximum time on the vehicle exceeds the direct travel time but not more than 30% + 3 min. In both DRT and FRT, we minimized the total travel time of the vehicles, assuming that the number of available vehicles and their capacity are not binding and that the number of bus stops per route is limited to 1, 2, 3, or 4. FRT routes are designed to serve all 288 addresses, assuming that these represent an example of a list of registered customers. For DRT, we evaluated the performance for batches of 20, 30, 40, and 50 train passengers, which we assume to be the range of batch sizes over different times of day. Each demand realization is a random sample drawn from the set of all 288 addresses. In total, we solved 320 DRT problem instances (four route lengths, four batch sizes, and 20 demand realizations for each batch size).

The dynamic programming algorithm was coded in Python 3.7 and the set covering model was solved by IBM Cplex 20.1.0. All experiments were conducted on an Intel Core i9-9900k (3.6Ghz) processor with 8 Cores, 3.6GHz computer with 64GB RAM, running on a 64-bit Windows 10 operating system.

The results in this section are organized as follows. Section 5.1 focuses on vehicle routing results, according to SOM1 and SOM2, for a given LOS. Section 5.2 addresses the implications of the LOS. Vehicle capacity estimates and passenger assignment implications are considered in Sect. 5.3.

5.1 Routing results for a given level of service

To examine the potential advantage of using flexible routes, we compare FRT and DRT feeder transportation systems, both under SOM1. Figures 2 and 3 illustrate the solutions for FRT and a representative DRT instance with 50 passengers. We also compare these with the option of TBS service. We focus on cost components of service operation, including fleet size and vehicle travel time. For the sake of completeness, we also report results related to the computational effort, especially the number of routes in the model. Note that in this section, we refer to the number of vehicles and the number of routes interchangeably, because we assume that the time between train arrivals at the station is long enough for all the vehicles to complete their routes.

The main results of this analysis are shown in Figs. 4 and 5. In Fig. 4, solid lines (denoted DRT20, DRT30, DRT40, and DRT50) represent the average number of vehicles (and routes) required to operate a DRT service for different batch sizes (20, 30, 40, and 50 passengers). The yellow dotted line (denoted DRT_MAX) shows the maximum number of routes needed over all demand realizations (obtained when the batch size is 50), which may be considered as the necessary estimated fleet size. The green dashed line (denoted FRT) represents the number of lines required to operate a fixed route transit (FRT) service, which is also the fleet size in this case.

As the figure shows, if routes can have up to four bus stops, a fleet of five vehicles is sufficient, which is clearly much better than the 17 vehicles needed if routes are restricted to one bus stop. However, the four-stop case is only marginally better than the six vehicles needed if routes are restricted to three stops. The figure also shows the advantage of DRT over FRT, which requires seven vehicles with four-stop routes, i.e., a 40% difference.

The average number of routes (shown in solid lines) also decreases with the number of stops, reaching values of 2.8–4.1 routes if four stops are allowed. The pattern is similar to the maximum number of vehicles, i.e., the marginal difference decreases quickly. If batch sizes by time of day can be reasonably predicted, the expected number of routes as a function of batch size can be used to plan driver shifts and thus reduce costs even further, compared to FRT.

Finally, note that even if single-stop shared taxi DRT is used, the average number of routes needed to serve batches of 20, 30, 40, and 50 passengers is about 10, 12, 13, and 15, respectively, thus reducing substantially the number of vehicles required by a TBS where a separate vehicle is required for each passenger.

Recall that under SOM1, the number of vehicles (and routes) that are used in the solution is not constrained and is not optimized, as our objective function is to minimize the total travel times of the vehicles. To understand the tradeoff between

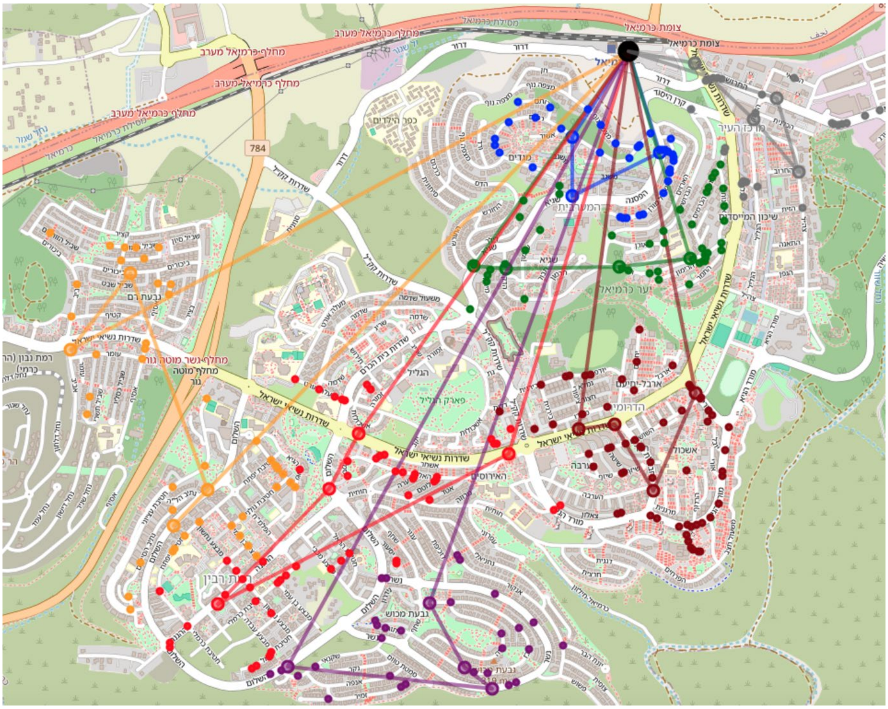


Fig. 2 Example of an FRT solution serving all registered passengers

the number of vehicles and the total travel time, we resolved all 320 instances using the SOM2 model, where the number of vehicles was limited to be one less than the number obtained using the SOM1 model. Interestingly, none of the constrained instances was feasible, which implies that the solution of the SOM1 model may simultaneously minimize the travel time and the number of vehicles.

We believe that the number of vehicles is likely minimized by the SOM1 model due to the special structure of the time windows in the problem. Recall that the allowed arrival time at each bus stop grows with its distance from the depot. This time window structure results in optimal routes that visit the bus stop at an approximately increasing distance from the depot. Therefore, the return to the depot from the last (farthest) stop is relatively costly in terms of the travel time objective of SOM1. Therefore, a model that minimizes travel time is also likely to minimize the number of routes and, thus, the number of vehicles.

Figure 5 focuses on variable costs and presents the optimal *average vehicle travel time per passenger* (AVTTPP). As expected, Fig. 5 shows that the AVTTPP decreases as the number of bus stops per route and the number of passenger requests per batch increase. We note that, with a TBS service, the AVTTPP is 28.7 min. According to the AVTTPP measure: DRT is always much better than TBS; four-stop DRT is substantially better than one-stop DRT; and, most importantly, for a given number of stops per route DRT is much better than FRT (1.8-fold for 40 passenger batches and four-stop routes). The direction in these three comparisons is obvious,

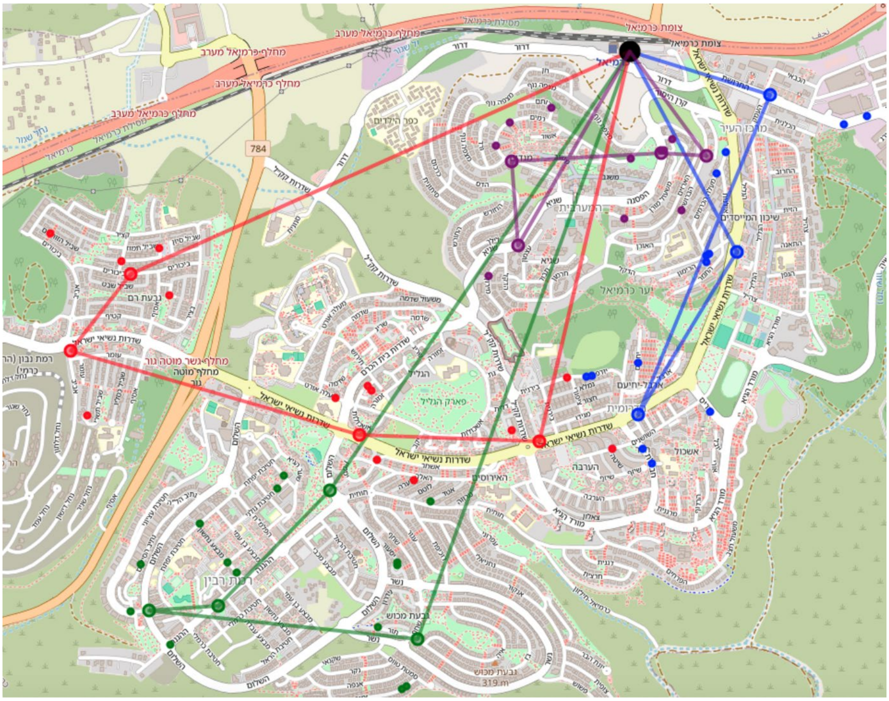


Fig. 3 Example of a DRT solution for one batch of 50 passengers

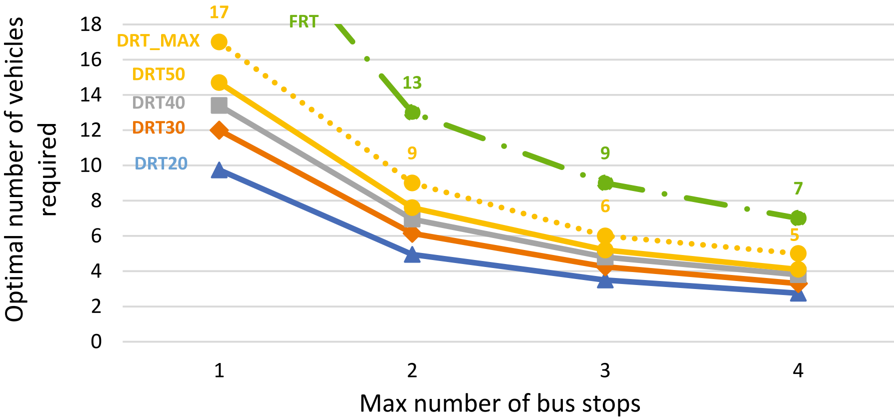


Fig. 4 Number of vehicles required under SOM1

but the results show that the magnitude of the DRT advantage is very substantial, and may justify the additional complexity of operating a DRT service rather than FRT.

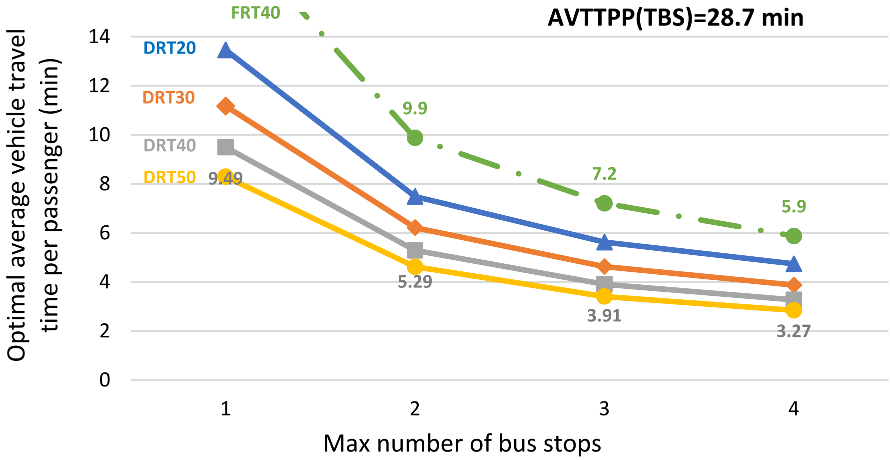


Fig. 5 Average vehicle travel time per passenger (AVTTPP) for DRT and FRT as a function of batch size

Similar to the case of the fixed costs (Fig. 4), the marginal benefit of additional stops on AVTTPP diminishes rather quickly, although the improvement when increasing the maximal number of bus stops from three to four is not negligible, e.g., 20% from 3.91 to 3.27 in the case of 40 passenger batches.

Figure 6 illustrates the economies of scale of the DRT operation by showing part of the results of Fig. 5 while focusing on four-stop routes. The AVTTPP, when the batch size is 50, is 40% lower compared to a batch size of 20. Therefore, if two (or more) operators share the same market, lower efficiency should be expected, and thus, competition within a specific train station may be detrimental to system cost.

While the focus of this paper is on service design and not on computational issues, for the sake of completeness, we present a brief discussion of key computational aspects. Figure 7 shows the number of non-dominated routes and the average number of relevant routes per batch on a logarithmic scale. As expected, the number of routes increases nearly exponentially with the number of stops and reaches almost 10 million non-dominated routes for up to four bus stops, which took us about 3 min to generate. However, under a specific demand realization, the number of routes is reduced by almost two orders of magnitude, and thus, the size of the input data for the set covering problem is also significantly reduced. With the procedure described in Sect. 4.2, the input data for each of these set covering problems were generated in about 3 min on average. The resulting set covering problems were solved by Cplex in about a minute on average. The computation time once the requests are known was on average four minutes and never exceeded 5.5 min.

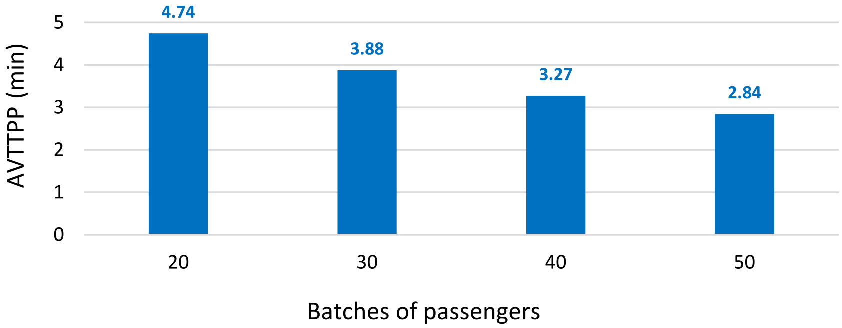


Fig. 6 Economics of scale in terms of average vehicle travel time per passenger (AVTTPP) for DRT with routes of up to four bus stops

5.2 Level of service analyses

This section deals with the role of LOS constraints in our model. We begin with an inspection of the solutions presented in Sect. 5.1 and examine how tight the LOS constraints are. Then, we consider another set of experiments with tighter LOS constraints, and discuss their effect on DRT resources (average vehicle travel time and number of vehicles required in the fleet). Finally, we observe the effect of LOS constraints on the computational effort.

Recall that the LOS in our DRT model is defined by upper bounds on walking times and on excess in-vehicle ride-time. Walking times are discussed in Sect. 5.3, while in this section, we focus on excess ride times, which is the difference between the actual ride time to a given bus stop along a chosen route and the direct travel time from the depot to the same stop. In our DRT model, the excess ride time is bounded by a sum of an additive constant and a fraction of the direct travel time. In the results presented in Sect. 5.1, the additive constant was 3 min, and the fraction was 30%.

To verify that the excess time constraint is binding in our example, we analyze the slack between the actual ride time and its upper bound. In Fig. 8, the circles represent ride times until each stop along every route in 20 demand realizations, when the batch size is 50 passengers, and the length of the routes is up to four bus stops. The actual ride time is presented on the vertical axis, while the direct travel time is presented on the horizontal axis. The circle colors represent the order of the stop along the route, and their size is proportional to the number of passengers dropped off at each specific stop along their routes. The 45-degree green line is obviously the lower bound on the actual ride time, and the red line shows the upper bound dictated by the required LOS (up to 3 min and 30% excess time).

We observe that 31.9% of the passengers arrive at their drop-off stop directly, while 19.7% of the passengers arrive at their drop-off stop not earlier than 2 min before the upper bound. That is, the constraint in this case is likely to be binding.

Figure 9 shows the cumulative distribution of the minimum differences between the maximum allowed and the actual ride time (referred to as “slack”)

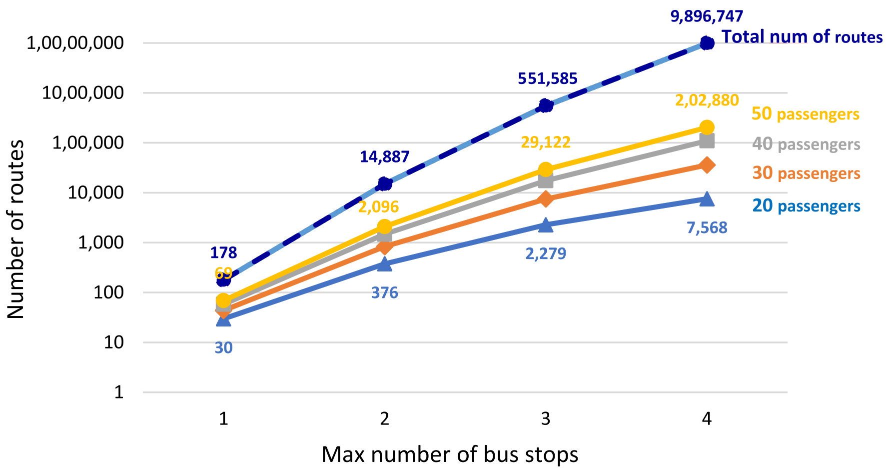


Fig. 7 Comparing the number of non-dominated routes and relevant routes

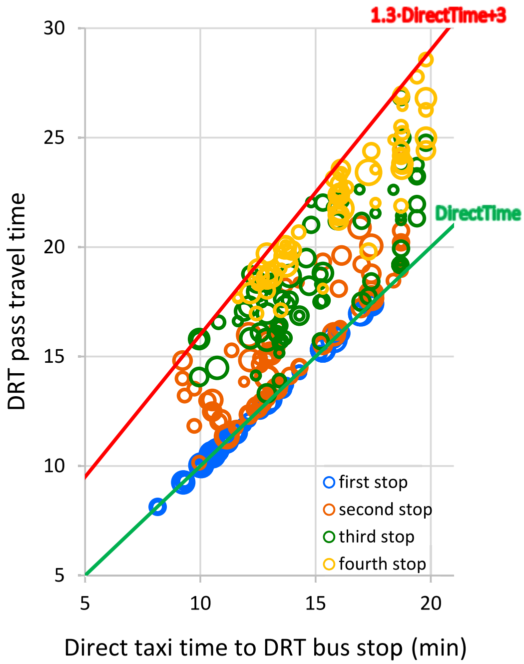
for the same 20 demand realizations of Fig. 8. We observe that in all examined realizations, at least one stop along one route satisfies the LOS constraint with a small margin of less than 2 min. In addition, if we tighten the constraint by a minute, the currently optimal solutions will violate the LOS constraint in 70% of these realizations.

The next set of analyses compares two levels of service in terms of excess ride time: A) 3min + $1.3\times c_{i0}$ ("low LOS") and B) 2min + $1.2\times c_{i0}$ ("high LOS"). In both cases, the maximum allowed walking time is 6 min. Figure 10 compares the optimal average vehicle travel time per passenger for the two LOSs. As expected, with high LOS, where the constraints on the ride times of the passengers to the desired destination are more demanding, the average vehicle travel time per passenger increases, implying additional variable costs to the operator. Additionally, as the number of stops on a route increases, the effect increases, reaching 6.7% on routes with up to four bus stops, since the strict service-level constraint eliminates even more of the existing long routes with low LOS. As expected, with direct routes (one stop), there are no differences. Analyses of this type can help regulators to make the appropriate tradeoff between LOS and operational costs.

We examined in addition the impact of LOS on the number of vehicles and found it to be modest. To be specific, the largest impact was observed in the case of 50 passengers and routes with up to four bus stops, and in these instances, the number of routes per batch increased on average only marginally, by 0.25 vehicles.

The LOS constraints also affect computational effort. The number of non-dominated routes with up to two, three and four bus stops decreased, respectively, by 20% (from 14,887 to 11,926), 53% (from 551,585 to 259,944) and 73% (from 9,896,747 to 2,654,049). The decrease in the number of routes after filtering irrelevant routes to a specific batch is similar. For example, on routes with up to four bus stops, the reduction is about 70% for all batch sizes. As a result, the problems become easier to solve by the procedure described in Sect. 4.2.

Fig. 8 Distribution of ride times of DRT passengers, for routes up to four bus stops, when the batch size is 50 passengers



The input data for each of the set covering problems were generated in about 1 min on average (three times faster than the low LOS case). The resulting set covering problems were solved by Cplex in about half a minute on average (two times faster than the low LOS case). The computation time - once the requests were known - was on average 1.5 min and never exceeded 2 min.

5.3 Vehicle capacity estimates

Decisions about vehicle capacities can be made in many ways. Recall that in Sect. 5.1, we examined the number of vehicles, and since driver labor is a dominant cost factor, we assume that vehicle capacity is a secondary consideration. Operators may be given the option to determine vehicle capacities, including the possibility of a mixed fleet consisting of vehicles with different capacities, as part of their operational responsibility. Our focus is on the design stage, when a regulator is interested in capacity estimates, to the extent that these impact initial estimates of costs. In this context, we assume that all vehicles have the same capacity, and we offer a way to estimate the minimal required capacity under this assumption.

For vehicle capacity estimates, an additional assumption is needed regarding the assignment of passengers to vehicles. We consider two modes of operation: a) every passenger is free to choose among all the vehicles that serve a bus stop within a given walking time from their destination (SOM1); or b) the operator assigns passengers to specific vehicles in a way that minimizes the maximal needed vehicle

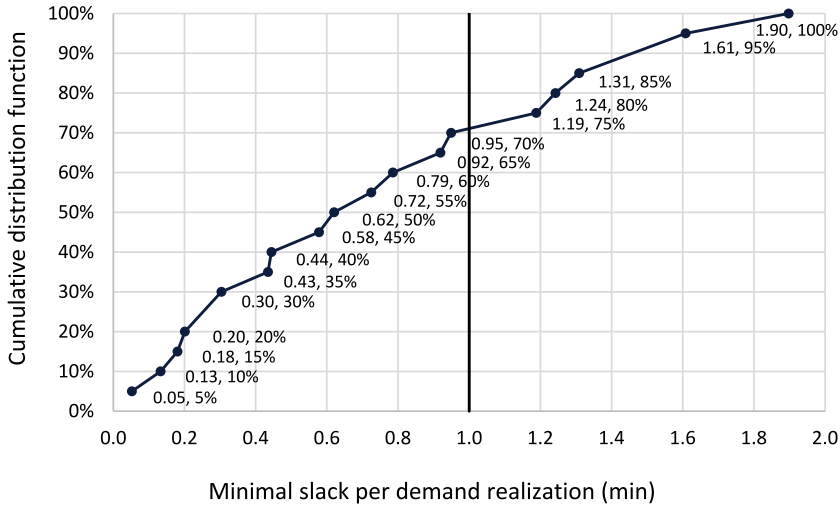


Fig. 9 Cumulative distribution of the minimal slack in 20 demand realizations (batch size = 50, up to four stops per route)

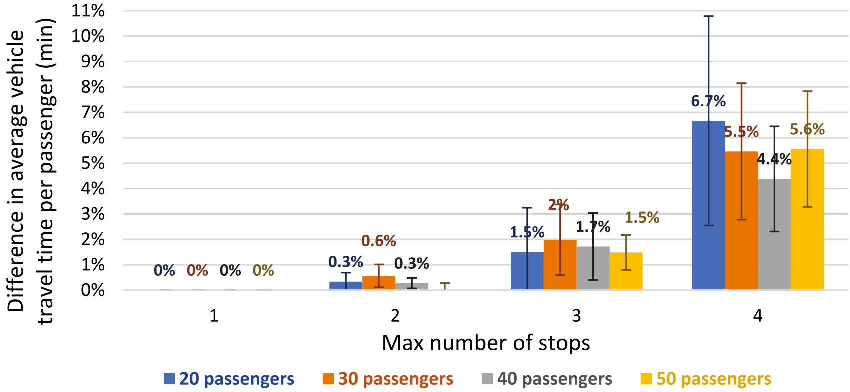


Fig. 10 The difference in optimal average vehicle travel time per passenger between LOS **A** and LOS **B**

capacity (SOM3). These two options span the range of required vehicle capacity from the worst-case (SOM1) to the best-case (SOM3) scenario, from the operator's perspective.

A difference between the two options may occur only if some "between vehicle flexibility" exists, as defined in Sect. 4.3 (5). We present first a verification that our test case exhibits such flexibility. We then show capacity estimates for DRT (under SOM1 and SOM3) and for FRT (under SOM1). Finally, we examine the difference between SOM1 and SOM3 from the point of view of passenger LOS, especially in terms of walking times.

Table 4 shows the values of between route flexibility rates. As expected, the flexibility rate increases with batch size. However, with an increase in the number of stops in the route, “between routes flexibility” usually decreases. This can be explained by the fact that with more stops per route, the choice of a passenger between several relevant bus stops may be within the same route, thus implying “within route flexibility” rather than “between route flexibility”.

Given that flexibilities exist in our test case, we now consider their impact on vehicle occupancies under SOM1 and SOM3. Figures 11a and b show the cumulative distribution of maximal occupancy values for all demand realizations for the largest batch sizes of 40 and 50 passengers, respectively, when the length of the routes is up to four bus stops. Each line in these figures represents the cumulative distribution on maximal occupancies under three different assumptions: SOM1 for DRT in blue; SOM3 for DRT in orange; and FRT (under SOM1) in green.

The 100th percentile of the maximal occupancy, for 50 passengers demand realizations (see Fig. 11.b) is 24 for SOM1, 21 for SOM3, and 20 for FRT. We note that the 100th percentile is not representative, and therefore, the design should consider the proportion of demand realizations that can be served optimally within a given vehicle capacity, while assuming that if this proportion is reasonable, various ad-hoc operational fixes can be implemented. These fixes may include outsourcing the service for the remaining passengers, solving the capacity-constrained routing problem (possibly heuristically), etc.

Figure 11 shows that the optimal routing solution for SOM1 with up to four bus stops and 50 passengers per batch (respectively, 40 passengers) can be implemented using 20-seat vehicles in 85% (respectively, 95%) of the demand realizations of the respective sizes. Under SOM3, these proportions increase to 90% and 100%, respectively. In conclusion, a vehicle capacity of 20 seems reasonably sufficient according to these results.

When reducing the capacity to 15 seats, the percentage of demand realizations in which the operator can implement the optimal vehicle routing solution drops to 10% and 60%, for batch sizes of 50 and 40, respectively, under SOM1, and 30% and 65%, respectively, under SOM3. Operating the service with 15-seat vehicles while meeting the required LOS may therefore be challenging, and could require more vehicles and drivers, as well as a possibly longer distance of vehicle travel.

Figure 11 also shows that in FRT, the occupancies are slightly lower than in DRT. Recall that as shown in Sect. 5.1 FRT service requires substantially more vehicles (i.e., more lines) and substantially longer traveled distances than DRT.

Passenger assignment option (SOM3 vs. SOM1) may affect individual LOS in terms of walking times and arrival times. We compare the walking times under SOM3 and under SOM1, assuming that when given the choice, passengers select a route and bus stop to minimize their walking time. In the case of 50 passengers with routes of up to four bus stops, walking times increase for 16 passengers (out of a total of 1,000 passengers), where ten of them experience an additional walking time of more than 1 min, and the maximum addition is 3.17 min. To put these values in perspective, under these conditions, average walking times are around 4 min (227–259 s) for DRT, and around 2 min (115–127 s) for TBS. If we assume that passengers minimize their arrival time to their destination (including the walk from the bus stop to the destination), the impact of passenger assignment according to SOM3 on arrival times is of similar magnitude. In the case of 50 passengers with routes of up to four bus stops, arrival times increase

Table 4 Flexibility rate by batch size and number of bus stops per route

Max number of bus stops Level demand	1	2	3	4
20	8.5%	3.8%	1.8%	2.5%
30	9.8%	4.8%	2.0%	2.7%
40	12.4%	6.3%	4.1%	3.6%
50	15.1%	6.7%	5.9%	4.6%

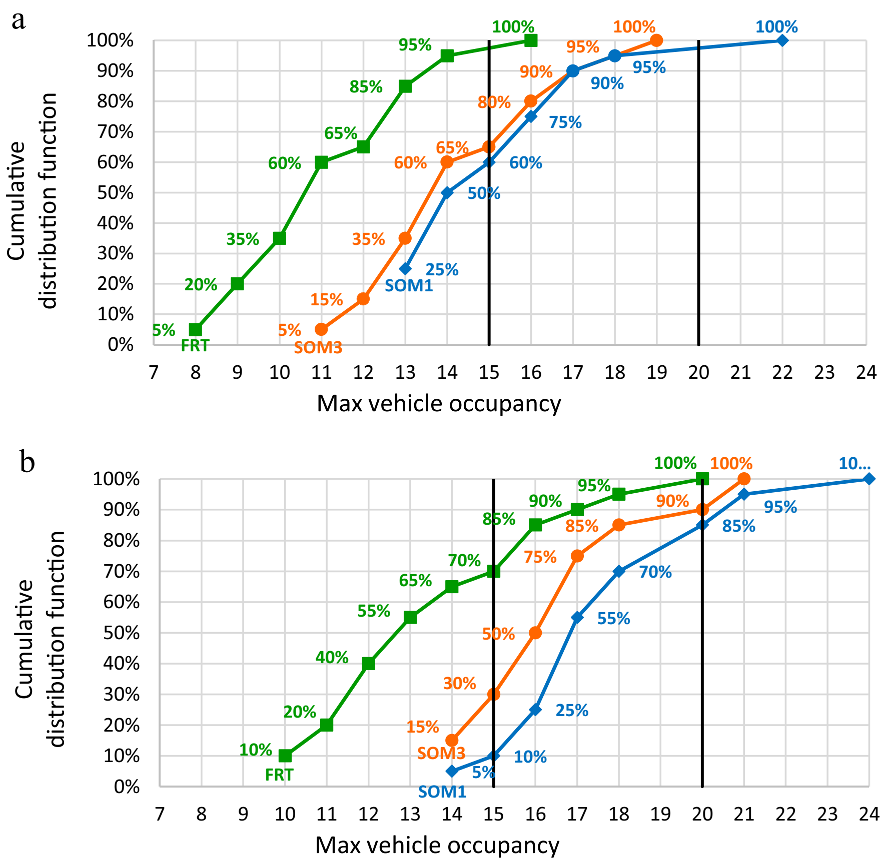


Fig. 11 **a** Cumulative distribution of maximal occupancy, for routes up to four bus stops, when the batch size is 40 passengers. **b** Cumulative distribution of maximal occupancy, for routes up to four bus stops, when the batch size is 50 passengers

for 12 out of 1000 passengers, while 4 of them arrive 4 min later, and the maximal delay in arrival time is 8 min. These differences in walking times and arrival times may be noticeable from a personal perspective, even though the number of meaningfully affected passengers is very small.

6 Conclusions and future research

The goal of this research was to offer quantitative methods for the design of partially flexible DRT services. Our focus was shared first-and-last-mile transit systems (SFLMTS), primarily at rural areas where fixed route services may be less efficient. We assume that the feeder service is associated with an infrequent long-haul service (e.g., by train), which determines the schedule.

A feeder service should cover passenger needs in both directions, to and from the train station. Service design considerations in the two directions are rather similar, and the associated optimization problems are completely equivalent. For convenience of the exposition, we addressed the train alighting scenario, under three simplified operational models. Our main formulation, the DRT 1-M problem, addresses vehicle routing while ensuring a high LOS in terms of walking time and excess ride time for all passengers. A unique feature of this formulation is the consideration of several possible drop-off points for each passenger, which is not a typical element in DRT research. We show that the model can be solved exactly (to optimality) for a realistic case study in fairly short computing times, and we demonstrate the usefulness of the model results for design purposes.

Our main conclusion is that shared DRT services can be very effective for serving last-mile connectivity from train stations by providing a guarantee for high service levels while balancing operational efficiency. These findings support our preliminary hypotheses about the match between the need for SFLMTS in rural and suburban areas, and the capabilities of DRT 1-M with optimized (or nearly optimized) routing in response to materialized demand. While many previous studies on DRT/MoD services considered various applications, the research in the context of SFLMTS has been relatively limited and should be explored more thoroughly.

Demand-responsive routes, when compared to providing the same LOS with fixed routes, enable a substantial reduction of fleet size (by approximately 30% in the peak demand case), as well as a reduction in vehicle travel time (by nearly 50%).

We found substantial economies of scale, illustrated by a 40% cost reduction when comparing batch sizes of 20 and 50. The design implication is that competition between multiple operators may be detrimental to system performance and benefits. We used the methods to examine the impact of modifying the LOS on operating costs (in terms of vehicle travel time). We also showed that simplified models can be used to provide initial estimates of desired vehicle capacity. These estimates are helpful during design and procurement stages, while operational considerations may require refined assessment.

The main limitation of this study is the consideration of a specific case study. While the methods proposed here are likely to be applicable in other contexts, the specific estimates may vary by location according to demand profile and network topology. Therefore, future research may address the generalizability of our findings. Similar DRT services may also be useful in other settings, including airports, schools, industrial parks, universities, and community centers. Evaluating such settings is also an interesting option for additional research.

There are several possible extensions to the models explored here. For example, we may consider multiple batches of passengers on both directions (inbound and outbound) from several trains over a certain planning horizon. Obviously, the SOMs proposed here abstract out important operational aspects of the service. Therefore, it is interesting to evaluate the usefulness of these SOMs to support service design decisions. One way to accomplish this evaluation is to develop simulation tools that consider travel time uncertainties, cancelations, and other user choices.

Appendix

Appendix A: Routing algorithm correctness

In Sect. 4, we presented two formulations for the DRT-1M problem, an arc-index formulation (1) and a set covering formulation (2). If the set covering problem considers all feasible routes, then the two formulations are clearly equivalent, as is the case in many other VRPs (Toth and Vigo, 2014). The equivalence still holds if all non-dominated feasible routes are considered. In the case of ties, that is, if multiple routes with the same cost cover the same subset of customer demands, considering only one of them is enough. Therefore, to show the correctness of the routing algorithm presented in Sect. 4, and that indeed formulations (1) and (2) are equivalent in our context, we need to prove Theorem 1.

Theorem 1: *Every route $r \in \mathcal{R}$ is feasible, the cost c_r computed in Algorithm 2 for every route $r \in \mathcal{R}$ is correct, and given any alternative feasible (candidate) route $\hat{r} \notin \mathcal{R}$ serving the $\hat{k}$ bus stops in $B_{\hat{r}}$ at a total cost of $c_{\hat{r}}$, there exists a route $r \in \mathcal{R}$, such that $B_r = B_{\hat{r}}$ and $c_r \leq c_{\hat{r}}$.*

The full proof of Theorem 1 is provided here for the sake of completeness, as well as since it clarifies the details of the DP method, including the connection between DP states, open routes, and closed routes. To prove Theorem 1, we will use the following additional notation. For a given route, r, serving k bus stops, let $b_r[k']$ for $0 \leq k' \leq k+1$ be the k' stop in the sequence, where the first and last stops must be in the depot, and thus, $b_r[0] = b_r[k+1] = 0$. We denote the partial set of the first k' bus stops in r by $B_r(k')$ and the cost of the sequence until the k' stop by $c_r(k')$, where $c_r(0) = 0$; $c_r(k'+1) = c_r(k') + c_{b_r[k'],b_r[k'+1]}$; and $c_r(k+1) = c_r$ is the full cost of the closed route. A route is feasible if $k \leq K$ (satisfied by all $r \in \mathcal{R}$), and if

$$c_r(k') \leq \varphi^{\max}_{b_r[k']} \forall 1 \leq k' \leq k.$$

The proof is divided into three parts, Lemma 1 deals with the correctness of Algorithm 1, while Lemma 2 and Lemma 3 deal with the correctness of Algorithm 2.

Lemma 1: *For every* $1 \leq k \leq K$ *and every feasible candidate route* $\hat{r} \notin \mathcal{R}$ *of length not less than* k, *where the last bus stop is* $j = b_{\hat{r}}[k]$, *there exists a state* $(B,j) \in \Omega_k$, *from which the route* r (*not necessarily in* $\mathcal{R}$), *which is constructed by the trace variables*, *is feasible*, *such that the following conditions hold*:

1A. $B = B_r = B_{\hat{r}}$

1B. $C(B,j) = c_r(k) \leq \varphi_j^{\max}$

1C. $C(B,j) \leq c_{\hat{r}}(k)$.

Proof: the proof is by induction (which in principle could start from zero). In the case of $k = 1$, let $j = b_{\hat{r}}(1)$ and $B = \{j\}$. For every bus stop $j \in S$, we know that $\varphi_j^{\max} \geq c_{0j}$, and hence, the condition in line 6 of Algorithm 1 is valid and, therefore, $(\{j\},j) \in \Omega_1$ and $C(\{j\},j) = c_{0j} = c_{\hat{r}}(1)$. The single-stop route $[0,j,0]$ and $B_r = B = \{j\}$ is feasible. Thus, all three conditions (1A, 1B, 1C) hold.

Assume by induction that the conditions hold for $k-1$. Let $j = b_{\hat{r}}(k)$, $j_1 = b_{\hat{r}}(k-1)$, $B = B_{\hat{r}}$, and $B_1 = B_{\hat{r}}(k-1)$. By the assumption of the induction $(B_1,j_1) \in \Omega_{k-1}$ and there exist a feasible route r_1 of length $k-1$, such that r_1 is the route constructed by the trace variables from the state (B_1,j_1), and in addition, $B_{r_1} = B_1 = B_{\hat{r}}(k-1)$; $C(B_1,j_1) = c_{r_1}(k-1) \leq \varphi_{j_1}^{\max}$; $C(B_1,j_1) \leq c_{\hat{r}}(k-1)$. We will examine the route r which extends r_1 by appending bus stop j at the end, right before the depot (as stop k). Consider the situation when the loop in line 2 of Algorithm 1 reaches k and the loop in line 4 reaches the state (B_1,j_1). Since $\hat{r}$ is feasible, $c_{\hat{r}}(k) = c_{\hat{r}}(k-1) + c_{j_1 j} \leq \varphi_j^{\max}$. Therefore, $C(B_1,j_1) + c_{j_1 j} \leq \varphi_j^{\max}$ and the condition in line 6 holds. Assume first that line 7 holds; thus, the state (B,j) is added to Ω_k in line 8, line 9 ensures that $C(B,j) = C(B_1,j_1) + c_{j_1 j} = c_{\hat{r}}(k-1) + c_{j_1 j} = c_{\hat{r}}(k)$, and line 10 sets $j^*(B,j) = j_1$. If we assume further that the values of $C(B,j)$ and $j^*(B,j)$ do not change later, then all three conditions hold. If line 7 does not hold, or if the values $C(B,j)$ and $j^*(B,j)$ change later, we still have that $(B,j) \in \Omega_k$ and that there is another route r' with lower cost for which all the conditions are met. QED.

Lemma 2: *For all chosen routes* $r \in \mathcal{R}$ (*of length* $1 \leq k \leq K$), *the feasibility condition* $c_r(k') \leq \varphi_{b_r[k']}^{\max}$ *holds for all* $1 \leq k' \leq k$ *and* c_r *as computed by Algorithm* 2 *equals* $c_r(k+1)$.

Proof: according to line 14 of Algorithm 2, every route $r \in \mathcal{R}$ of length k is created from an element $B \in \Omega$ which is associated by the trace variable $j = j_0^*(B)$ to a state $(B,j) \in \Omega_k$. By Lemma 1 the route r constructed by the trace variables from (B,j) satisfies the feasibility conditions, and the cost $C(B,j) = c_r(k)$ is correct. The value of c_r is set in line 15 to be equal to $C_0(B)$, which is determined in line 6 or 10 of Algorithm 2 to be $C(B,j) + c_{j0} = c_r(k) + c_{j0} = c_r(k+1)$. This shows the correctness of the cost calculation. QED.

Lemma 3: *For any feasible candidate route* $\hat{r} \notin \mathcal{R}$, *there exists a chosen route* $r \in \mathcal{R}$, *such that* $B_r = B_{\hat{r}}$ *and* $c_r \leq c_{\hat{r}}$.

Proof: Let $k = |B_{\hat{r}}|$ be the length of route $\hat{r}$, let $B = B_{\hat{r}}$, and let $j = b_{\hat{r}}[k]$. By Lemma 1, there exists a state $(B, j) \in \Omega_k$, with $C(B, j) \leq c_{\hat{r}}(k)$. The loop in line 3 of Algorithm 2 ensures that for all $(B, j) \in \Omega_k$, there will be an element $B \in \Omega$, with $C_0(B) \leq C(B, j) + c_{j0} \leq c_{\hat{r}}(k+1) = c_{\hat{r}}$. Lemma 2 shows that for each element $B \in \Omega$, Algorithm 2 constructs a feasible non-dominated route $r \in \mathcal{R}$, such that $c_r = C_0(B) \leq c_{\hat{r}}$. QED.

Conclusion: The combination of Lemma 2 and Lemma 3 proves Theorem 1. QED.

Acknowledgements This work was supported by: Institute for Innovative Transportation and Capsule, Tel Aviv University; The Israeli Center for Promoting Research in Smart Transportation; Ben-Gurion University student fellowship; Ben-Gurion University of the Negev Students Assistance Fund; Sami Shamoon College of Engineering Research and Development Authority Fund.

Funding Open access funding provided by Sami Shamoon College of Engineering. The first author was partially supported by the Israeli Smart Transportation Research Center (ISTRC).

Data availability The data used in this study are available upon request; please contact the corresponding author to obtain them.

Declarations

Conflict of interest The authors report that there are no competing interests to declare.

References

Avermann N, Schlüter J (2019) Determinants of customer satisfaction with a true door-to-door DRT service in rural Germany. Res Transp Bus Manag 32:100420. https://doi.org/10.1016/j.rtbm.2019.100420

Bansal P, Liu Y, Daziano R, Samaranayake S (2019) Impact of discerning reliability preferences of riders on the demand for mobility-on-demand services. Transp Lett 12(10):677–681. https://doi.org/10.1080/19427867.2019.1691298

Basu R, Araldo A, Akkinepally AP, Nahmias Biran BH, Basak K, Seshadri R, Deshmukh N, Kumar N, Azevedo CL, Ben-Akiva M (2018) Automated mobility-on-demand vs. mass transit: a multi-modal activity-driven agent-based simulation approach. Transp Res Rec 2672(8):608–618. https://doi.org/10.1177/0361198118758630

Ceder, A. (2015) Public transit planning and operation: Modeling, practice and behavior (2nd ed.). Boca Raton: CRC Press. https://doi.org/10.1201/b18689.

Charisis A, Iliopoulou C, Kepaptsoglou K (2018) DRT route design for the first/last mile problem: model and application to Athens, Greece. Public Transp 10(3):499–527. https://doi.org/10.1007/s12469-018-0188-0

Corvello V, Musmanno R, Pavone G, Santoro F, Vocaturo F (2023) Optimization and public transport tendering: a case study in Southern Italy. Public Transp 15:697–719. https://doi.org/10.1007/s12469-023-00324-9

Czioska P, Kutadinata R, Trifunović A, Winter S, Sester M, Friedrich B (2019) Real-world meeting points for shared demand-responsive transportation systems. Public Transp 11:341–377. https://doi.org/10.1007/s12469-019-00207-y

Errico F, Crainic TG, Malucelli F, Nonato M (2013) A survey on planning semi-flexible transit systems: methodological issues and a unifying framework. Transp Res Part C: Emerg Technol 36:324–338. https://doi.org/10.1016/j.trc.2013.08.010

Godachevich J, Tirachini A (2021) Does the measured performance of bus operators depend on the index chosen to assess reliability in contracts? An analysis of bus headway variability. Res Transp Econ 90:101000. https://doi.org/10.1016/j.retrec.2020.101000

Leffler D, Burghout W, Jenelius E, Cats O (2021) Simulation of fixed versus on-demand station-based feeder operations. Transp Res Part C: Emerg Technol 132:103401. https://doi.org/10.1016/j.trc.2021.103401

Lu Y, Kimpton A, Prato C-G, Sipe N, Corcoran J (2024) First and last mile travel mode choice: a systematic review of the empirical literature. Int J Sustain Transp 18(1):1–14. https://doi.org/10.1080/15568318.2023.2218285

Ma TY, Rasulkhani S, Chow JYJ, Klein S (2018) An integrated dynamic ridesharing dispatch and idle vehicle repositioning strategy on a bimodal transport network. Transp Res Arena, Vienna, Austria. https://doi.org/10.5281/zenodo.2155709

Markov I, Guglielmetti R, Laumanns M, Fernández-Antolín A, de Souza R (2021) Simulation-based design and analysis of on-demand mobility services. Transp Res Part A 149:170–205. https://doi.org/10.1016/j.tra.2021.01.004

Mehran B, Yang Y, Mishra S (2020) Analytical models for comparing operational costs of regular bus and semi-flexible transit services. Public Transp 12(1):147–169. https://doi.org/10.1007/s12469-019-00222-z

Miller C, Tucker A, Zemlin R (1960) Integer programming formulations and traveling salesman problems. J Assoc Comput Mach 7:326–329. https://doi.org/10.1145/321043.321046

Mishra S, Mehran B (2023) Effect of stochastic vehicle arrival and passenger demand on semi-flexible transit design. Public Transp 15:721–745. https://doi.org/10.1007/s12469-023-00325-8

Montenegro BDG, Sörensen K, Vansteenwegen P (2020) A demand-responsive feeder system with mandatory and optional, clustered bus-stops. Working papers 2020006, University of Antwerp, Faculty of Business and Economics. https://repository.uantwerpen.be/docman/irua/e8dc5e/rps_2020_006.pdf, Accessed 11 Aug 2025

Montenegro BDG, Sörensen K, Vansteenwegen P (2021) A large neighborhood search algorithm to optimize a demand-responsive feeder service. Transp Res Part C: Emerg Technol 127:103102. https://doi.org/10.1016/j.trc.2021.103102

Navidi Z, Ronald N, Winter S (2018) Comparison between ad-hoc demand responsive and conventional transit: a simulation study. Public Transp 10(1):147–167. https://doi.org/10.1007/s12469-017-0173-z

Perera S, Ho C, Hensher D (2020) Resurgence of demand responsive transit services – Insights from BRIDJ trials in Inner West of Sydney. Australia Res Transp Econom 83:100904. https://doi.org/10.1016/j.retrec.2020.100904

Raghunathan AU, Bergman D, Hooker J, Serra T, Kobori S (2018a) Seamless multimodal transportation scheduling. arXiv preprint arXiv:1807.09676. https://doi.org/10.48550/arXiv.1807.09676

Raghunathan AU, Bergman D, Hooker JN, Serra T, Kobori S (2018b) The integrated last-mile transportation problem (ILMTP). ICAPS. https://doi.org/10.1609/icaps.v28i1.13917

Sangveraphunsiri T, Cassidy MJ, Daganzo CF (2022) Jitney-lite: a flexible-route feeder service for developing countries. Transp Res Part B: Methodol 156:1–13. https://doi.org/10.1016/j.trb.2021.12.015

Scheltes A, de Almeida Correia GH (2017) Exploring the use of automated vehicles as last mile connection of train trips through an agent-based simulation model: an application to Delft, Netherlands. Proceed Int J Transp Sci Technol 6:28–41. https://doi.org/10.1016/j.ijtst.2017.05.004

Shaheen S, Chan N (2016) Mobility and the sharing economy: potential to facilitate the first- and last-mile public transit connections. Built Environ 42(4):573–588. https://doi.org/10.2148/benv.42.4.573

Shehadeh KS, Wang H, Zhang P (2021) Fleet sizing and allocation for on-demand last-mile transportation systems. Transp Res Part C: Emerg Technol 132:103387. https://doi.org/10.1016/j.trc.2021.103387

Shen Y, Zhang H, Zhao J (2018) Integrating shared autonomous vehicle in public transportation system: a supply-side simulation of the first-mile service in Singapore. Transp Res Part A: Policy Pract 113:125–136. https://doi.org/10.1016/j.tra.2018.04.004

Sörensen L, Bossert A, Jokinen J, Schlüter J (2021) How much flexibility does rural public transport need? – Implications from a fully flexible DRT system. Transp Policy 100:5–20. https://doi.org/10.1016/j.tranpol.2020.09.005

Sun B, Wei M, Zhu S (2018) Optimal design of demand-responsive feeder transit services with passengers' multiple time windows and satisfaction. Future Internet 10(30):3. https://doi.org/10.3390/fi10030030

Tachet R, Sagarra O, Santi P, Resta G, Szell M, Strogatz SH, Ratti C (2017) Scaling law of urban ride sharing. Sci Rep 7:42868. https://doi.org/10.1038/srep42868

Toth P, Vigo D (2014) Vehicle Routing: Problems, Methods, and Applications. MOS-SIAM Series Optimization, Philadelphia. https://doi.org/10.1137/1.9781611973594.fm

Vansteenwegen P, Melis L, Aktaş D, Montenegro BDG, Vieira FS, Sörensen K (2022) A survey on demand-responsive public bus systems. Transp Res Part C: Emerg Technol 137:103573. https://doi.org/10.1016/j.trc.2022.103573

Wang H (2017) Routing and scheduling for a last-mile transportation system. Transport Sci 53(1):131–147. https://doi.org/10.1287/trsc.2017.0753

Wang Z, Yu J, Hao W, Chen T, Wang Y (2020) Designing high freedom responsive feeder transit system with multitype vehicles. J Adv Tansp 2020:8365194. https://doi.org/10.1155/2020/8365194

Wang Z, Yu J, Hao W, Xiang J (2021) Joint optimization of running route and scheduling for the mixed demand responsive feeder transit with time-dependent travel times. IEEE Trans Intell Transp Syst 22(4):2498–2509. https://doi.org/10.1109/TITS.2020.3041743

Yang R, Lai M (2020) Multi-level analyses on the nearest-first matching policy of on-demand chauffeured ride-hailing service. Int J Sustain Transp 15:749–767. https://doi.org/10.1080/15568318.2020.1795760